

12

Orthogonal Functions and Fourier Series

EXERCISES 12.1

Orthogonal Functions

1. $\int_{-2}^2 x x^2 dx = \frac{1}{4} x^4 \Big|_{-2}^2 = 0$
2. $\int_{-1}^1 x^3 (x^2 + 1) dx = \frac{1}{6} x^6 \Big|_{-1}^1 + \frac{1}{4} x^4 \Big|_{-1}^1 = 0$
3. $\int_0^2 e^x (x e^{-x} - e^{-x}) dx = \int_0^2 (x - 1) dx = \left(\frac{1}{2} x^2 - x \right) \Big|_0^2 = 0$
4. $\int_0^\pi \cos x \sin^2 x \, dx = \frac{1}{3} \sin^3 x \Big|_0^\pi = 0$
5. $\int_{-\pi/2}^{\pi/2} x \cos 2x \, dx = \frac{1}{2} \left(\frac{1}{2} \cos 2x + x \sin 2x \right) \Big|_{-\pi/2}^{\pi/2} = 0$
6. $\int_{\pi/4}^{5\pi/4} e^x \sin x \, dx = \left(\frac{1}{2} e^x \sin x - \frac{1}{2} e^x \cos x \right) \Big|_{\pi/4}^{5\pi/4} = 0$
7. For $m \neq n$

$$\begin{aligned}
 & \int_0^{\pi/2} \sin(2n+1)x \sin(2m+1)x \, dx \\
 &= \frac{1}{2} \int_0^{\pi/2} (\cos 2(n-m)x - \cos 2(n+m+1)x) \, dx \\
 &= \frac{1}{4(n-m)} \sin 2(n-m)x \Big|_0^{\pi/2} - \frac{1}{4(n+m+1)} \sin 2(n+m+1)x \Big|_0^{\pi/2} = 0.
 \end{aligned}$$

For $m = n$

$$\begin{aligned}\int_0^{\pi/2} \sin^2(2n+1)x \, dx &= \int_0^{\pi/2} \left(\frac{1}{2} - \frac{1}{2} \cos 2(2n+1)x \right) dx \\ &= \frac{1}{2}x \Big|_0^{\pi/2} - \frac{1}{4(2n+1)} \sin 2(2n+1)x \Big|_0^{\pi/2} = \frac{\pi}{4}\end{aligned}$$

so that

$$\|\sin(2n+1)x\| = \frac{1}{2}\sqrt{\pi}.$$

8. For $m \neq n$

$$\begin{aligned}\int_0^{\pi/2} \cos(2n+1)x \cos(2m+1)x \, dx \\ &= \frac{1}{2} \int_0^{\pi/2} (\cos 2(n-m)x + \cos 2(n+m+1)x) \, dx \\ &= \frac{1}{4(n-m)} \sin 2(n-m)x \Big|_0^{\pi/2} + \frac{1}{4(n+m+1)} \sin 2(n+m+1)x \Big|_0^{\pi/2} = 0.\end{aligned}$$

For $m = n$

$$\begin{aligned}\int_0^{\pi/2} \cos^2(2n+1)x \, dx &= \int_0^{\pi/2} \left(\frac{1}{2} + \frac{1}{2} \cos 2(2n+1)x \right) dx \\ &= \frac{1}{2}x \Big|_0^{\pi/2} + \frac{1}{4(2n+1)} \sin 2(2n+1)x \Big|_0^{\pi/2} = \frac{\pi}{4}\end{aligned}$$

so that

$$\|\cos(2n+1)x\| = \frac{1}{2}\sqrt{\pi}.$$

9. For $m \neq n$

$$\begin{aligned}\int_0^{\pi} \sin nx \sin mx \, dx &= \frac{1}{2} \int_0^{\pi} (\cos(n-m)x - \cos(n+m)x) \, dx \\ &= \frac{1}{2(n-m)} \sin(n-m)x \Big|_0^{\pi} - \frac{1}{2(n+m)} \sin(n+m)x \Big|_0^{\pi} = 0.\end{aligned}$$

For $m = n$

$$\int_0^{\pi} \sin^2 nx \, dx = \int_0^{\pi} \left(\frac{1}{2} - \frac{1}{2} \cos 2nx \right) dx = \frac{1}{2}x \Big|_0^{\pi} - \frac{1}{4n} \sin 2nx \Big|_0^{\pi} = \frac{\pi}{2}$$

so that

$$\|\sin nx\| = \sqrt{\frac{\pi}{2}}.$$

10. For $m \neq n$

$$\begin{aligned}\int_0^p \sin \frac{n\pi}{p}x \sin \frac{m\pi}{p}x \, dx &= \frac{1}{2} \int_0^p \left(\cos \frac{(n-m)\pi}{p}x - \cos \frac{(n+m)\pi}{p}x \right) dx \\ &= \frac{p}{2(n-m)\pi} \sin \frac{(n-m)\pi}{p}x \Big|_0^p - \frac{p}{2(n+m)\pi} \sin \frac{(n+m)\pi}{p}x \Big|_0^p = 0.\end{aligned}$$

For $m = n$

$$\int_0^p \sin^2 \frac{n\pi}{p}x \, dx = \int_0^p \left(\frac{1}{2} - \frac{1}{2} \cos \frac{2n\pi}{p}x \right) dx = \frac{1}{2}x \Big|_0^p - \frac{p}{4n\pi} \sin \frac{2n\pi}{p}x \Big|_0^p = \frac{p}{2}$$

12.1 Orthogonal Functions

so that

$$\left\| \sin \frac{n\pi}{p} x \right\| = \sqrt{\frac{p}{2}}.$$

11. For $m \neq n$

$$\begin{aligned} \int_0^p \cos \frac{n\pi}{p} x \cos \frac{m\pi}{p} x \, dx &= \frac{1}{2} \int_0^p \left(\cos \frac{(n-m)\pi}{p} x + \cos \frac{(n+m)\pi}{p} x \right) dx \\ &= \frac{p}{2(n-m)\pi} \sin \frac{(n-m)\pi}{p} x \Big|_0^p + \frac{p}{2(n+m)\pi} \sin \frac{(n+m)\pi}{p} x \Big|_0^p = 0. \end{aligned}$$

For $m = n$

$$\int_0^p \cos^2 \frac{n\pi}{p} x \, dx = \int_0^p \left(\frac{1}{2} + \frac{1}{2} \cos \frac{2n\pi}{p} x \right) dx = \frac{1}{2} x \Big|_0^p + \frac{p}{4n\pi} \sin \frac{2n\pi}{p} x \Big|_0^p = \frac{p}{2}.$$

Also

$$\int_0^p 1 \cdot \cos \frac{n\pi}{p} x \, dx = \frac{p}{n\pi} \sin \frac{n\pi}{p} x \Big|_0^p = 0 \quad \text{and} \quad \int_0^p 1^2 \, dx = p$$

so that

$$\|1\| = \sqrt{p} \quad \text{and} \quad \left\| \cos \frac{n\pi}{p} x \right\| = \sqrt{\frac{p}{2}}.$$

12. For $m \neq n$, we use Problems 11 and 10:

$$\begin{aligned} \int_{-p}^p \cos \frac{n\pi}{p} x \cos \frac{m\pi}{p} x \, dx &= 2 \int_0^p \cos \frac{n\pi}{p} x \cos \frac{m\pi}{p} x \, dx = 0 \\ \int_{-p}^p \sin \frac{n\pi}{p} x \sin \frac{m\pi}{p} x \, dx &= 2 \int_0^p \sin \frac{n\pi}{p} x \sin \frac{m\pi}{p} x \, dx = 0. \end{aligned}$$

Also

$$\int_{-p}^p \sin \frac{n\pi}{p} x \cos \frac{m\pi}{p} x \, dx = \frac{1}{2} \int_{-p}^p \left(\sin \frac{(n-m)\pi}{p} x + \sin \frac{(n+m)\pi}{p} x \right) dx = 0,$$

$$\int_{-p}^p 1 \cdot \cos \frac{n\pi}{p} x \, dx = \frac{p}{n\pi} \sin \frac{n\pi}{p} x \Big|_{-p}^p = 0,$$

$$\int_{-p}^p 1 \cdot \sin \frac{n\pi}{p} x \, dx = -\frac{p}{n\pi} \cos \frac{n\pi}{p} x \Big|_{-p}^p = 0,$$

and

$$\int_{-p}^p \sin \frac{n\pi}{p} x \cos \frac{n\pi}{p} x \, dx = \int_{-p}^p \frac{1}{2} \sin \frac{2n\pi}{p} x \, dx = -\frac{p}{4n\pi} \cos \frac{2n\pi}{p} x \Big|_{-p}^p = 0.$$

For $m = n$

$$\int_{-p}^p \cos^2 \frac{n\pi}{p} x \, dx = \int_{-p}^p \left(\frac{1}{2} + \frac{1}{2} \cos \frac{2n\pi}{p} x \right) dx = p,$$

$$\int_{-p}^p \sin^2 \frac{n\pi}{p} x \, dx = \int_{-p}^p \left(\frac{1}{2} - \frac{1}{2} \cos \frac{2n\pi}{p} x \right) dx = p,$$

and

$$\int_{-p}^p 1^2 \, dx = 2p$$

so that

$$\|1\| = \sqrt{2p}, \quad \left\| \cos \frac{n\pi}{p} x \right\| = \sqrt{p}, \quad \text{and} \quad \left\| \sin \frac{n\pi}{p} x \right\| = \sqrt{p}.$$

13. Since

$$\int_{-\infty}^{\infty} e^{-x^2} \cdot 1 \cdot 2x \, dx = -e^{-x^2} \Big|_{-\infty}^0 - e^{-x^2} \Big|_0^{\infty} = 0,$$

$$\begin{aligned} \int_{-\infty}^{\infty} e^{-x^2} \cdot 1 \cdot (4x^2 - 2) \, dx &= 2 \int_{-\infty}^{\infty} x (2xe^{-x^2}) \, dx - 2 \int_{-\infty}^{\infty} e^{-x^2} \, dx \\ &= 2 \left(-xe^{-x^2} \Big|_{-\infty}^{\infty} + \int_{-\infty}^{\infty} e^{-x^2} \, dx \right) - 2 \int_{-\infty}^{\infty} e^{-x^2} \, dx \\ &= 2 \left(-xe^{-x^2} \Big|_{-\infty}^0 - xe^{-x^2} \Big|_0^{\infty} \right) = 0, \end{aligned}$$

and

$$\begin{aligned} \int_{-\infty}^{\infty} e^{-x^2} \cdot 2x \cdot (4x^2 - 2) \, dx &= 4 \int_{-\infty}^{\infty} x^2 (2xe^{-x^2}) \, dx - 4 \int_{-\infty}^{\infty} xe^{-x^2} \, dx \\ &= 4 \left(-x^2 e^{-x^2} \Big|_{-\infty}^{\infty} + 2 \int_{-\infty}^{\infty} xe^{-x^2} \, dx \right) - 4 \int_{-\infty}^{\infty} xe^{-x^2} \, dx \\ &= 4 \left(-x^2 e^{-x^2} \Big|_{-\infty}^0 - x^2 e^{-x^2} \Big|_0^{\infty} \right) + 2 \int_{-\infty}^{\infty} 2xe^{-x^2} \, dx = 0, \end{aligned}$$

the functions are orthogonal.

14. Since

$$\int_0^{\infty} e^{-x} \cdot 1(1-x) \, dx = (x-1)e^{-x} \Big|_0^{\infty} - \int_0^{\infty} e^{-x} \, dx = 0,$$

$$\begin{aligned} \int_0^{\infty} e^{-x} \cdot 1 \cdot \left(\frac{1}{2}x^2 - 2x + 1 \right) \, dx &= \left(2x - 1 - \frac{1}{2}x^2 \right) e^{-x} \Big|_0^{\infty} + \int_0^{\infty} e^{-x}(x-2) \, dx \\ &= 1 + (2-x)e^{-x} \Big|_0^{\infty} + \int_0^{\infty} e^{-x} \, dx = 0, \end{aligned}$$

and

$$\begin{aligned} \int_0^{\infty} e^{-x} \cdot (1-x) \left(\frac{1}{2}x^2 - 2x + 1 \right) \, dx &= \int_0^{\infty} e^{-x} \left(-\frac{1}{2}x^3 + \frac{5}{2}x^2 - 3x + 1 \right) \, dx \\ &= e^{-x} \left(\frac{1}{2}x^3 - \frac{5}{2}x^2 + 3x - 1 \right) \Big|_0^{\infty} + \int_0^{\infty} e^{-x} \left(-\frac{3}{2}x^2 + 5x - 3 \right) \, dx \\ &= 1 + e^{-x} \left(\frac{3}{2}x^2 - 5x + 3 \right) \Big|_0^{\infty} + \int_0^{\infty} e^{-x}(5-3x) \, dx \\ &= 1 - 3 + e^{-x}(3x-5) \Big|_0^{\infty} - 3 \int_0^{\infty} e^{-x} \, dx = 0, \end{aligned}$$

the functions are orthogonal.

15. By orthogonality $\int_a^b \phi_0(x)\phi_n(x)dx = 0$ for $n = 1, 2, 3, \dots$; that is, $\int_a^b \phi_n(x)dx = 0$ for $n = 1, 2, 3, \dots$.

12.1 Orthogonal Functions

16. Using the facts that ϕ_0 and ϕ_1 are orthogonal to ϕ_n for $n > 1$, we have

$$\begin{aligned}\int_a^b (\alpha x + \beta) \phi_n(x) dx &= \alpha \int_a^b x \phi_n(x) dx + \beta \int_a^b 1 \cdot \phi_n(x) dx \\ &= \alpha \int_a^b \phi_1(x) \phi_n(x) dx + \beta \int_a^b \phi_0(x) \phi_n(x) dx \\ &= \alpha \cdot 0 + \beta \cdot 0 = 0\end{aligned}$$

for $n = 2, 3, 4, \dots$.

17. Using the fact that ϕ_n and ϕ_m are orthogonal for $n \neq m$ we have

$$\begin{aligned}\|\phi_m(x) + \phi_n(x)\|^2 &= \int_a^b [\phi_m(x) + \phi_n(x)]^2 dx = \int_a^b [\phi_m^2(x) + 2\phi_m(x)\phi_n(x) + \phi_n^2(x)] dx \\ &= \int_a^b \phi_m^2(x) dx + 2 \int_a^b \phi_m(x)\phi_n(x) dx + \int_a^b \phi_n^2(x) dx \\ &= \|\phi_m(x)\|^2 + \|\phi_n(x)\|^2.\end{aligned}$$

18. Setting

$$0 = \int_{-2}^2 f_3(x) f_1(x) dx = \int_{-2}^2 (x^2 + c_1 x^3 + c_2 x^4) dx = \frac{16}{3} + \frac{64}{5} c_2$$

and

$$0 = \int_{-2}^2 f_3(x) f_2(x) dx = \int_{-2}^2 (x^3 + c_1 x^4 + c_2 x^5) dx = \frac{64}{5} c_1$$

we obtain $c_1 = 0$ and $c_2 = -5/12$.

19. Since $\sin nx$ is an odd function on $[-\pi, \pi]$,

$$(1, \sin nx) = \int_{-\pi}^{\pi} \sin nx dx = 0$$

and $f(x) = 1$ is orthogonal to every member of $\{\sin nx\}$. Thus $\{\sin nx\}$ is not complete.

20. $(f_1 + f_2, f_3) = \int_a^b [f_1(x) + f_2(x)] f_3(x) dx = \int_a^b f_1(x) f_3(x) dx + \int_a^b f_2(x) f_3(x) dx = (f_1, f_3) + (f_2, f_3)$

21. (a) The fundamental period is $2\pi/2\pi = 1$.

(b) The fundamental period is $2\pi/(4/L) = \frac{1}{2}\pi L$.

(c) The fundamental period of $\sin x + \sin 2x$ is 2π .

(d) The fundamental period of $\sin 2x + \cos 4x$ is $2\pi/2 = \pi$.

(e) The fundamental period of $\sin 3x + \cos 4x$ is 2π since the smallest integer multiples of $2\pi/3$ and $2\pi/4 = \pi/2$ that are equal are 3 and 4, respectively.

(f) The fundamental period of $f(x)$ is $2\pi/(n\pi/p) = 2p/n$.

22. (a) Following the pattern established by $\phi_1(x)$ and $\phi_2(x)$ we have

$$\phi_3(x) = f_3(x) - \frac{(f_3, \phi_0)}{(\phi_0, \phi_0)} \phi_0(x) - \frac{(f_3, \phi_1)}{(\phi_1, \phi_1)} \phi_1(x) - \frac{(f_3, \phi_2)}{(\phi_2, \phi_2)} \phi_2(x).$$

- (b) To show mutual orthogonality we compute (ϕ_0, ϕ_1) , (ϕ_0, ϕ_2) , and (ϕ_1, ϕ_2) using properties (i), (ii), and (iii) from this section in the text.

$$\begin{aligned}(\phi_0, \phi_1) &= \left(\phi_0, f_1 - \frac{(f_1, \phi_0)}{(\phi_0, \phi_0)} \phi_0 \right) = (\phi_0, f_1) - \frac{(f_1, \phi_0)}{(\phi_0, \phi_0)} (\phi_0, \phi_0) = (\phi_0, f_1) - (f_1, \phi_0) = 0 \\(\phi_0, \phi_2) &= \left(\phi_0, f_2 - \frac{(f_2, \phi_0)}{(\phi_0, \phi_0)} \phi_0 - \frac{(f_2, \phi_1)}{(\phi_1, \phi_1)} \phi_1 \right) = (\phi_0, f_2) - \frac{(f_2, \phi_0)}{(\phi_0, \phi_0)} (\phi_0, \phi_0) - \frac{(f_2, \phi_1)}{(\phi_1, \phi_1)} (\phi_0, \phi_1) \\&= (\phi_0, f_2) - (f_2, \phi_0) - 0 = 0 \\(\phi_1, \phi_2) &= \left(\phi_1, f_2 - \frac{(f_2, \phi_0)}{(\phi_0, \phi_0)} \phi_0 - \frac{(f_2, \phi_1)}{(\phi_1, \phi_1)} \phi_1 \right) = (\phi_1, f_2) - \frac{(f_2, \phi_0)}{(\phi_0, \phi_0)} (\phi_1, \phi_0) - \frac{(f_2, \phi_1)}{(\phi_1, \phi_1)} (\phi_1, \phi_1) \\&= (\phi_1, f_2) - 0 - (f_2, \phi_1) = 0.\end{aligned}$$

23. (a) First we identify $f_0(x) = 1$, $f_1(x) = x$, $f_2(x) = x^2$, and $f_3(x) = x^3$. Then, we use the formulas from Problem 22. First, we have $\phi_0(x) = f_0(x) = 1$. Then

$$(f_1, \phi_0) = (x, 1) = \int_{-1}^1 x \, dx = 0 \quad \text{and} \quad (\phi_0, \phi_0) = \int_{-1}^1 1 \, dx = 2,$$

so

$$\phi_1(x) = f_1(x) - \frac{(f_1, \phi_0)}{(\phi_0, \phi_0)} \phi_0(x) = x - \frac{0}{2}(1) = x.$$

Next

$$(f_2, \phi_0) = (x^2, 1) = \int_{-1}^1 x^2 \, dx = \frac{2}{3}, \quad (f_2, \phi_1) = (x^2, x) = \int_{-1}^1 x^3 \, dx = 0, \quad \text{and} \quad (\phi_1, \phi_1) = \int_{-1}^1 x^2 \, dx = \frac{2}{3},$$

so

$$\phi_2(x) = f_2(x) - \frac{(f_2, \phi_0)}{(\phi_0, \phi_0)} \phi_0(x) - \frac{(f_2, \phi_1)}{(\phi_1, \phi_1)} \phi_1(x) = x^2 - \frac{2/3}{2}(1) - \frac{0}{2/3}(x) = x^2 - \frac{1}{3}.$$

Finally,

$$(f_3, \phi_0) = (x^3, 1) = \int_{-1}^1 x^3 \, dx = 0, \quad (f_3, \phi_1) = (x^3, x) = \int_{-1}^1 x^4 \, dx = \frac{2}{5},$$

and

$$(f_3, \phi_2) = \left(x^3, x^2 - \frac{1}{3} \right) = \int_{-1}^1 \left(x^5 - \frac{1}{3} x^3 \right) dx = 0,$$

so

$$\phi_3(x) = f_3(x) - \frac{(f_3, \phi_0)}{(\phi_0, \phi_0)} \phi_0(x) - \frac{(f_3, \phi_1)}{(\phi_1, \phi_1)} \phi_1(x) - \frac{(f_3, \phi_2)}{(\phi_2, \phi_2)} \phi_2(x) = x^3 - 0 - \frac{2/5}{2/3}(x) - 0 = x^3 - \frac{3}{5}x.$$

- (b) Recall from Section 5.3 that the first four Legendre polynomials are $P_0(x) = 1$, $P_1(x) = x$, $P_2(x) = \frac{3}{2}x^2 - \frac{1}{2}$, and $P_3(x) = \frac{5}{2}x^3 - \frac{3}{2}x$. We then see that $\phi_0(x) = P_0(x)$, $\phi_1(x) = P_1(x)$, $\phi_2(x) = x^2 - \frac{1}{3} = \frac{2}{3}(\frac{3}{2}x^2 - \frac{1}{2}) = \frac{2}{3}P_2(x)$, and $\phi_3(x) = x^3 - \frac{3}{5}x = \frac{2}{5}(\frac{5}{2}x^3 - \frac{3}{2}x) = \frac{2}{5}P_3(x)$.

24. (i): $(f_1, f_2) = \int_a^b f_1(x)f_2(x)dx = \int_a^b f_2(x)f_1(x)dx = (f_2, f_1).$

(ii): $(kf_1, f_2) = \int_a^b kf_1(x)f_2(x)dx = k \int_a^b f_1(x)f_2(x)dx = k(f_1, f_2).$

(iii): If $f_1(x) = 0$ then $(f_1, f_1) = \int_a^b 0 \, dx = 0$; if $f_1(x) \neq 0$ then $(f_1, f_1) = \int_a^b [f_1(x)]^2 dx > 0$ since $[f_1(x)]^2 > 0$.

(iv): $(f_1 + f_2, f_3) = \int_a^b [f_1(x) + f_2(x)]f_3(x)dx = \int_a^b [f_1(x)f_3(x) + f_2(x)f_3(x)]dx$
 $= \int_a^b f_1(x)f_3(x)dx + \int_a^b f_2(x)f_3(x)dx = (f_1, f_3) + (f_2, f_3).$

12.1 Orthogonal Functions

25. In R^3 the set $\{\mathbf{i}, \mathbf{j}\}$ is not complete since $\mathbf{k}$ is orthogonal to both $\mathbf{i}$ and $\mathbf{j}$. The set $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ is complete. To see this suppose that $a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$ is orthogonal to $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$. Then

$$0 = (a\mathbf{i} + b\mathbf{j} + c\mathbf{k}, \mathbf{i}) = a(\mathbf{i}, \mathbf{i}) + b(\mathbf{j}, \mathbf{i}) + c(\mathbf{k}, \mathbf{i}) = a(1) + b(0) + c(0) = a.$$

Similarly, $b = 0$ and $c = 0$. Thus, the only vector in R^3 orthogonal to $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ is $\mathbf{0}$, so $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ is complete.

EXERCISES 12.2

Fourier Series

1. $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} 1 dx = 1$
 $a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos \frac{n\pi}{\pi} x dx = \frac{1}{\pi} \int_0^{\pi} \cos nx dx = 0$
 $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin \frac{n\pi}{\pi} x dx = \frac{1}{\pi} \int_0^{\pi} \sin nx dx = \frac{1}{n\pi} (1 - \cos n\pi) = \frac{1}{n\pi} [1 - (-1)^n]$

$$f(x) = \frac{1}{2} + \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n} \sin nx$$
2. $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^0 (-1) dx + \frac{1}{\pi} \int_0^{\pi} 2 dx = 1$
 $a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_{-\pi}^0 -\cos nx dx + \frac{1}{\pi} \int_0^{\pi} 2 \cos nx dx = 0$
 $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_{-\pi}^0 -\sin nx dx + \frac{1}{\pi} \int_0^{\pi} 2 \sin nx dx = \frac{3}{n\pi} [1 - (-1)^n]$

$$f(x) = \frac{1}{2} + \frac{3}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n} \sin nx$$
3. $a_0 = \int_{-1}^1 f(x) dx = \int_{-1}^0 1 dx + \int_0^1 x dx = \frac{3}{2}$
 $a_n = \int_{-1}^1 f(x) \cos n\pi x dx = \int_{-1}^0 \cos n\pi x dx + \int_0^1 x \cos n\pi x dx = \frac{1}{n^2\pi^2} [(-1)^n - 1]$
 $b_n = \int_{-1}^1 f(x) \sin n\pi x dx = \int_{-1}^0 \sin n\pi x dx + \int_0^1 x \sin n\pi x dx = -\frac{1}{n\pi}$

$$f(x) = \frac{3}{4} + \sum_{n=1}^{\infty} \left[\frac{(-1)^n - 1}{n^2\pi^2} \cos n\pi x - \frac{1}{n\pi} \sin n\pi x \right]$$
4. $a_0 = \int_{-1}^1 f(x) dx = \int_0^1 x dx = \frac{1}{2}$
 $a_n = \int_{-1}^1 f(x) \cos n\pi x dx = \int_0^1 x \cos n\pi x dx = \frac{1}{n^2\pi^2} [(-1)^n - 1]$
 $b_n = \int_{-1}^1 f(x) \sin n\pi x dx = \int_0^1 x \sin n\pi x dx = \frac{(-1)^{n+1}}{n\pi}$

- $$f(x) = \frac{1}{4} + \sum_{n=1}^{\infty} \left[\frac{(-1)^n - 1}{n^2 \pi^2} \cos n\pi x + \frac{(-1)^{n+1}}{n\pi} \sin n\pi x \right]$$
5. $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} x^2 dx = \frac{1}{3} \pi^2$
- $$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_0^{\pi} x^2 \cos nx dx = \frac{1}{\pi} \left(\frac{x^2}{\pi} \sin nx \Big|_0^{\pi} - \frac{2}{n} \int_0^{\pi} x \sin nx dx \right) = \frac{2(-1)^n}{n^2}$$
- $$b_n = \frac{1}{\pi} \int_0^{\pi} x^2 \sin nx dx = \frac{1}{\pi} \left(-\frac{x^2}{n} \cos nx \Big|_0^{\pi} + \frac{2}{n} \int_0^{\pi} x \cos nx dx \right) = \frac{\pi}{n} (-1)^{n+1} + \frac{2}{n^3 \pi} [(-1)^n - 1]$$
- $$f(x) = \frac{\pi^2}{6} + \sum_{n=1}^{\infty} \left[\frac{2(-1)^n}{n^2} \cos nx + \left(\frac{\pi}{n} (-1)^{n+1} + \frac{2[(-1)^n - 1]}{n^3 \pi} \right) \sin nx \right]$$
6. $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^0 \pi^2 dx + \frac{1}{\pi} \int_0^{\pi} (\pi^2 - x^2) dx = \frac{5}{3} \pi^2$
- $$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_{-\pi}^0 \pi^2 \cos nx dx + \frac{1}{\pi} \int_0^{\pi} (\pi^2 - x^2) \cos nx dx$$
- $$= \frac{1}{\pi} \left(\frac{\pi^2 - x^2}{n} \sin nx \Big|_0^{\pi} + \frac{2}{n} \int_0^{\pi} x \sin nx dx \right) = \frac{2}{n^2} (-1)^{n+1}$$
- $$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_{-\pi}^0 \pi^2 \sin nx dx + \frac{1}{\pi} \int_0^{\pi} (\pi^2 - x^2) \sin nx dx$$
- $$= \frac{\pi}{n} [(-1)^n - 1] + \frac{1}{\pi} \left(\frac{x^2 - \pi^2}{n} \cos nx \Big|_0^{\pi} - \frac{2}{n} \int_0^{\pi} x \cos nx dx \right) = \frac{\pi}{n} (-1)^n + \frac{2}{n^3 \pi} [1 - (-1)^n]$$
- $$f(x) = \frac{5\pi^2}{6} + \sum_{n=1}^{\infty} \left[\frac{2}{n^2} (-1)^{n+1} \cos nx + \left(\frac{\pi}{n} (-1)^n + \frac{2[1 - (-1)^n]}{n^3 \pi} \right) \sin nx \right]$$
7. $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} (x + \pi) dx = 2\pi$
- $$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_{-\pi}^{\pi} (x + \pi) \cos nx dx = 0$$
- $$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{2}{n} (-1)^{n+1}$$
- $$f(x) = \pi + \sum_{n=1}^{\infty} \frac{2}{n} (-1)^{n+1} \sin nx$$
8. $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} (3 - 2x) dx = 6$
- $$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_{-\pi}^{\pi} (3 - 2x) \cos nx dx = 0$$
- $$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} (3 - 2x) \sin nx dx = \frac{4}{n} (-1)^n$$
- $$f(x) = 3 + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \sin nx$$
9. $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} \sin x dx = \frac{2}{\pi}$
- $$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_0^{\pi} \sin x \cos nx dx = \frac{1}{2\pi} \int_0^{\pi} (\sin(1+n)x + \sin(1-n)x) dx$$

12.2 Fourier Series

$$= \frac{1 + (-1)^n}{\pi(1 - n^2)} \quad \text{for } n = 2, 3, 4, \dots$$

$$a_1 = \frac{1}{2\pi} \int_0^\pi \sin 2x \, dx = 0$$

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^\pi f(x) \sin nx \, dx = \frac{1}{\pi} \int_0^\pi \sin x \sin nx \, dx \\ &= \frac{1}{2\pi} \int_0^\pi (\cos(1-n)x - \cos(1+n)x) \, dx = 0 \quad \text{for } n = 2, 3, 4, \dots \end{aligned}$$

$$b_1 = \frac{1}{2\pi} \int_0^\pi (1 - \cos 2x) \, dx = \frac{1}{2}$$

$$f(x) = \frac{1}{\pi} + \frac{1}{2} \sin x + \sum_{n=2}^{\infty} \frac{1 + (-1)^n}{\pi(1 - n^2)} \cos nx$$

$$10. \quad a_0 = \frac{2}{\pi} \int_{-\pi/2}^{\pi/2} f(x) \, dx = \frac{2}{\pi} \int_0^{\pi/2} \cos x \, dx = \frac{2}{\pi}$$

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_{-\pi/2}^{\pi/2} f(x) \cos 2nx \, dx = \frac{2}{\pi} \int_0^{\pi/2} \cos x \cos 2nx \, dx = \frac{1}{\pi} \int_0^{\pi/2} (\cos(2n-1)x + \cos(2n+1)x) \, dx \\ &= \frac{2(-1)^{n+1}}{\pi(4n^2 - 1)} \end{aligned}$$

$$\begin{aligned} b_n &= \frac{2}{\pi} \int_{-\pi/2}^{\pi/2} f(x) \sin 2nx \, dx = \frac{2}{\pi} \int_0^{\pi/2} \cos x \sin 2nx \, dx = \frac{1}{\pi} \int_0^{\pi/2} (\sin(2n-1)x + \sin(2n+1)x) \, dx \\ &= \frac{4n}{\pi(4n^2 - 1)} \end{aligned}$$

$$f(x) = \frac{1}{\pi} + \sum_{n=1}^{\infty} \left[\frac{2(-1)^{n+1}}{\pi(4n^2 - 1)} \cos 2nx + \frac{4n}{\pi(4n^2 - 1)} \sin 2nx \right]$$

$$11. \quad a_0 = \frac{1}{2} \int_{-2}^2 f(x) \, dx = \frac{1}{2} \left(\int_{-1}^0 -2 \, dx + \int_0^1 1 \, dx \right) = -\frac{1}{2}$$

$$a_n = \frac{1}{2} \int_{-2}^2 f(x) \cos \frac{n\pi}{2} x \, dx = \frac{1}{2} \left(\int_{-1}^0 (-2) \cos \frac{n\pi}{2} x \, dx + \int_0^1 \cos \frac{n\pi}{2} x \, dx \right) = -\frac{1}{n\pi} \sin \frac{n\pi}{2}$$

$$b_n = \frac{1}{2} \int_{-2}^2 f(x) \sin \frac{n\pi}{2} x \, dx = \frac{1}{2} \left(\int_{-1}^0 (-2) \sin \frac{n\pi}{2} x \, dx + \int_0^1 \sin \frac{n\pi}{2} x \, dx \right) = \frac{3}{n\pi} \left(1 - \cos \frac{n\pi}{2} \right)$$

$$f(x) = -\frac{1}{4} + \sum_{n=1}^{\infty} \left[-\frac{1}{n\pi} \sin \frac{n\pi}{2} \cos \frac{n\pi}{2} x + \frac{3}{n\pi} \left(1 - \cos \frac{n\pi}{2} \right) \sin \frac{n\pi}{2} x \right]$$

$$12. \quad a_0 = \frac{1}{2} \int_{-2}^2 f(x) \, dx = \frac{1}{2} \left(\int_0^1 x \, dx + \int_1^2 1 \, dx \right) = \frac{3}{4}$$

$$a_n = \frac{1}{2} \int_{-2}^2 f(x) \cos \frac{n\pi}{2} x \, dx = \frac{1}{2} \left(\int_0^1 x \cos \frac{n\pi}{2} x \, dx + \int_1^2 \cos \frac{n\pi}{2} x \, dx \right) = \frac{2}{n^2\pi^2} \left(\cos \frac{n\pi}{2} - 1 \right)$$

$$\begin{aligned} b_n &= \frac{1}{2} \int_{-2}^2 f(x) \sin \frac{n\pi}{2} x \, dx = \frac{1}{2} \left(\int_0^1 x \sin \frac{n\pi}{2} x \, dx + \int_1^2 \sin \frac{n\pi}{2} x \, dx \right) \\ &= \frac{2}{n^2\pi^2} \left(\sin \frac{n\pi}{2} + \frac{n\pi}{2} (-1)^{n+1} \right) \end{aligned}$$

$$f(x) = \frac{3}{8} + \sum_{n=1}^{\infty} \left[\frac{2}{n^2\pi^2} \left(\cos \frac{n\pi}{2} - 1 \right) \cos \frac{n\pi}{2} x + \frac{2}{n^2\pi^2} \left(\sin \frac{n\pi}{2} + \frac{n\pi}{2} (-1)^{n+1} \right) \sin \frac{n\pi}{2} x \right]$$

13. $a_0 = \frac{1}{5} \int_{-5}^5 f(x) dx = \frac{1}{5} \left(\int_{-5}^0 1 dx + \int_0^5 (1+x) dx \right) = \frac{9}{2}$
 $a_n = \frac{1}{5} \int_{-5}^5 f(x) \cos \frac{n\pi}{5} x dx = \frac{1}{5} \left(\int_{-5}^0 \cos \frac{n\pi}{5} x dx + \int_0^5 (1+x) \cos \frac{n\pi}{5} x dx \right) = \frac{5}{n^2 \pi^2} [(-1)^n - 1]$
 $b_n = \frac{1}{5} \int_{-5}^5 f(x) \sin \frac{n\pi}{5} x dx = \frac{1}{5} \left(\int_{-5}^0 \sin \frac{n\pi}{5} x dx + \int_0^5 (1+x) \sin \frac{n\pi}{5} x dx \right) = \frac{5}{n\pi} (-1)^{n+1}$
 $f(x) = \frac{9}{4} + \sum_{n=1}^{\infty} \left[\frac{5}{n^2 \pi^2} [(-1)^n - 1] \cos \frac{n\pi}{5} x + \frac{5}{n\pi} (-1)^{n+1} \sin \frac{n\pi}{5} x \right]$
14. $a_0 = \frac{1}{2} \int_{-2}^2 f(x) dx = \frac{1}{2} \left(\int_{-2}^0 (2+x) dx + \int_0^2 2 dx \right) = 3$
 $a_n = \frac{1}{2} \int_{-2}^2 f(x) \cos \frac{n\pi}{2} x dx = \frac{1}{2} \left(\int_{-2}^0 (2+x) \cos \frac{n\pi}{2} x dx + \int_0^2 2 \cos \frac{n\pi}{2} x dx \right) = \frac{2}{n^2 \pi^2} [1 - (-1)^n]$
 $b_n = \frac{1}{2} \int_{-2}^2 f(x) \sin \frac{n\pi}{2} x dx = \frac{1}{2} \left(\int_{-2}^0 (2+x) \sin \frac{n\pi}{2} x dx + \int_0^2 2 \sin \frac{n\pi}{2} x dx \right) = \frac{2}{n\pi} (-1)^{n+1}$
 $f(x) = \frac{3}{2} + \sum_{n=1}^{\infty} \left[\frac{2}{n^2 \pi^2} [1 - (-1)^n] \cos \frac{n\pi}{2} x + \frac{2}{n\pi} (-1)^{n+1} \sin \frac{n\pi}{2} x \right]$
15. $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} e^x dx = \frac{1}{\pi} (e^{\pi} - e^{-\pi})$
 $a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{(-1)^n (e^{\pi} - e^{-\pi})}{\pi(1+n^2)}$
 $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_{-\pi}^{\pi} e^x \sin nx dx = \frac{(-1)^n n (e^{-\pi} - e^{\pi})}{\pi(1+n^2)}$
 $f(x) = \frac{e^{\pi} - e^{-\pi}}{2\pi} + \sum_{n=1}^{\infty} \left[\frac{(-1)^n (e^{\pi} - e^{-\pi})}{\pi(1+n^2)} \cos nx + \frac{(-1)^n n (e^{-\pi} - e^{\pi})}{\pi(1+n^2)} \sin nx \right]$
16. $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} (e^x - 1) dx = \frac{1}{\pi} (e^{\pi} - \pi - 1)$
 $a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_0^{\pi} (e^x - 1) \cos nx dx = \frac{[e^{\pi} (-1)^n - 1]}{\pi(1+n^2)}$
 $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_0^{\pi} (e^x - 1) \sin nx dx = \frac{1}{\pi} \left(\frac{n e^{\pi} (-1)^{n+1}}{1+n^2} + \frac{n}{1+n^2} + \frac{(-1)^n}{n} - \frac{1}{n} \right)$
 $f(x) = \frac{e^{\pi} - \pi - 1}{2\pi} + \sum_{n=1}^{\infty} \left[\frac{e^{\pi} (-1)^n - 1}{\pi(1+n^2)} \cos nx + \left(\frac{n}{1+n^2} [e^{\pi} (-1)^{n+1} + 1] + \frac{(-1)^n - 1}{n} \right) \sin nx \right]$

17. The function in Problem 5 is discontinuous at $x = \pi$, so the corresponding Fourier series converges to $\pi^2/2$ at $x = \pi$. That is,

$$\begin{aligned} \frac{\pi^2}{2} &= \frac{\pi^2}{6} + \sum_{n=1}^{\infty} \left[\frac{2(-1)^n}{n^2} \cos n\pi + \left(\frac{\pi}{n} (-1)^{n+1} + \frac{2[(-1)^n - 1]}{n^3 \pi} \right) \sin n\pi \right] \\ &= \frac{\pi^2}{6} + \sum_{n=1}^{\infty} \frac{2(-1)^n}{n^2} (-1)^n = \frac{\pi^2}{6} + \sum_{n=1}^{\infty} \frac{2}{n^2} = \frac{\pi^2}{6} + 2 \left(1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots \right) \end{aligned}$$

and

$$\frac{\pi^2}{6} = \frac{1}{2} \left(\frac{\pi^2}{2} - \frac{\pi^2}{6} \right) = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots$$

12.2 Fourier Series

At $x = 0$ the series converges to 0 and

$$0 = \frac{\pi^2}{6} + \sum_{n=1}^{\infty} \frac{2(-1)^n}{n^2} = \frac{\pi^2}{6} + 2 \left(-1 + \frac{1}{2^2} - \frac{1}{3^2} + \frac{1}{4^2} - \cdots \right)$$

so

$$\frac{\pi^2}{12} = 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots.$$

18. From Problem 17

$$\frac{\pi^2}{8} = \frac{1}{2} \left(\frac{\pi^2}{6} + \frac{\pi^2}{12} \right) = \frac{1}{2} \left(2 + \frac{2}{3^2} + \frac{2}{5^2} + \cdots \right) = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \cdots.$$

19. The function in Problem 7 is continuous at $x = \pi/2$ so

$$\frac{3\pi}{2} = f\left(\frac{\pi}{2}\right) = \pi + \sum_{n=1}^{\infty} \frac{2}{n} (-1)^{n+1} \sin \frac{n\pi}{2} = \pi + 2 \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots \right)$$

and

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots.$$

20. The function in Problem 9 is continuous at $x = \pi/2$ so

$$1 = f\left(\frac{\pi}{2}\right) = \frac{1}{\pi} + \frac{1}{2} + \sum_{n=2}^{\infty} \frac{1 + (-1)^n}{\pi(1 - n^2)} \cos \frac{n\pi}{2}$$

$$1 = \frac{1}{\pi} + \frac{1}{2} + \frac{2}{3\pi} - \frac{2}{3 \cdot 5\pi} + \frac{2}{5 \cdot 7\pi} - \cdots$$

and

$$\pi = 1 + \frac{\pi}{2} + \frac{2}{3} - \frac{2}{3 \cdot 5} + \frac{2}{5 \cdot 7} - \cdots$$

or

$$\frac{\pi}{4} = \frac{1}{2} + \frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \cdots.$$

21. Writing

$$f(x) = \frac{a_0}{2} + a_1 \cos \frac{\pi}{p} x + \cdots + a_n \cos \frac{n\pi}{p} x + \cdots + b_1 \sin \frac{\pi}{p} x + \cdots + b_n \sin \frac{n\pi}{p} x + \cdots$$

we see that $f^2(x)$ consists exclusively of squared terms of the form

$$\frac{a_0^2}{4}, \quad a_n^2 \cos^2 \frac{n\pi}{p} x, \quad b_n^2 \sin^2 \frac{n\pi}{p} x$$

and cross-product terms, with $m \neq n$, of the form

$$a_0 a_n \cos \frac{n\pi}{p} x, \quad a_0 b_n \sin \frac{n\pi}{p} x, \quad 2a_m a_n \cos \frac{m\pi}{p} x \cos \frac{n\pi}{p} x,$$

$$2a_m b_n \cos \frac{m\pi}{p} x \sin \frac{n\pi}{p} x, \quad 2b_m b_n \sin \frac{m\pi}{p} x \sin \frac{n\pi}{p} x.$$

The integral of each cross-product term taken over the interval $(-p, p)$ is zero by orthogonality. For the squared terms we have

$$\frac{a_0^2}{4} \int_{-p}^p dx = \frac{a_0^2 p}{2}, \quad a_n^2 \int_{-p}^p \cos^2 \frac{n\pi}{p} x dx = a_n^2 p, \quad b_n^2 \int_{-p}^p \sin^2 \frac{n\pi}{p} x dx = b_n^2 p.$$

Thus

$$RMS(f) = \sqrt{\frac{1}{4} a_0^2 + \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2)}.$$

EXERCISES 12.3

Fourier Cosine and Sine Series

1. Since $f(-x) = \sin(-3x) = -\sin 3x = -f(x)$, $f(x)$ is an odd function.
2. Since $f(-x) = -x \cos(-x) = -x \cos x = -f(x)$, $f(x)$ is an odd function.
3. Since $f(-x) = (-x)^2 - x = x^2 - x$, $f(x)$ is neither even nor odd.
4. Since $f(-x) = (-x)^3 + 4x = -(x^3 - 4x) = -f(x)$, $f(x)$ is an odd function.
5. Since $f(-x) = e^{|-x|} = e^{|x|} = f(x)$, $f(x)$ is an even function.
6. Since $f(-x) = e^{-x} - e^x = -f(x)$, $f(x)$ is an odd function.
7. For $0 < x < 1$, $f(-x) = (-x)^2 = x^2 = -f(x)$, $f(x)$ is an odd function.
8. For $0 \leq x < 2$, $f(-x) = -x + 5 = f(x)$, $f(x)$ is an even function.
9. Since $f(x)$ is not defined for $x < 0$, it is neither even nor odd.
10. Since $f(-x) = |(-x)^5| = |x^5| = f(x)$, $f(x)$ is an even function.
11. Since $f(x)$ is an odd function, we expand in a sine series:

$$b_n = \frac{2}{\pi} \int_0^\pi 1 \cdot \sin nx \, dx = \frac{2}{n\pi} [1 - (-1)^n].$$

Thus

$$f(x) = \sum_{n=1}^{\infty} \frac{2}{n\pi} [1 - (-1)^n] \sin nx.$$

12. Since $f(x)$ is an even function, we expand in a cosine series:

$$a_0 = \int_1^2 1 \, dx = 1$$

$$a_n = \int_1^2 \cos \frac{n\pi}{2} x \, dx = -\frac{2}{n\pi} \sin \frac{n\pi}{2}.$$

Thus

$$f(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{-2}{n\pi} \sin \frac{n\pi}{2} \cos \frac{n\pi}{2} x.$$

13. Since $f(x)$ is an even function, we expand in a cosine series:

$$a_0 = \frac{2}{\pi} \int_0^\pi x \, dx = \pi$$

$$a_n = \frac{2}{\pi} \int_0^\pi x \cos nx \, dx = \frac{2}{n^2\pi} [(-1)^n - 1].$$

Thus

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2\pi} [(-1)^n - 1] \cos nx.$$

12.3 Fourier Cosine and Sine Series

14. Since $f(x)$ is an odd function, we expand in a sine series:

$$b_n = \frac{2}{\pi} \int_0^\pi x \sin nx \, dx = \frac{2}{n}(-1)^{n+1}.$$

Thus

$$f(x) = \sum_{n=1}^{\infty} \frac{2}{n}(-1)^{n+1} \sin nx.$$

15. Since $f(x)$ is an even function, we expand in a cosine series:

$$a_0 = 2 \int_0^1 x^2 \, dx = \frac{2}{3}$$

$$a_n = 2 \int_0^1 x^2 \cos n\pi x \, dx = 2 \left(\frac{x^2}{n\pi} \sin n\pi x \Big|_0^1 - \frac{2}{n\pi} \int_0^1 x \sin n\pi x \, dx \right) = \frac{4}{n^2\pi^2}(-1)^n.$$

Thus

$$f(x) = \frac{1}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2\pi^2}(-1)^n \cos n\pi x.$$

16. Since $f(x)$ is an odd function, we expand in a sine series:

$$b_n = 2 \int_0^1 x^2 \sin n\pi x \, dx = 2 \left(-\frac{x^2}{n\pi} \cos n\pi x \Big|_0^1 + \frac{2}{n\pi} \int_0^1 x \cos n\pi x \, dx \right) = \frac{2(-1)^{n+1}}{n\pi} + \frac{4}{n^3\pi^3}[(-1)^n - 1].$$

Thus

$$f(x) = \sum_{n=1}^{\infty} \left(\frac{2(-1)^{n+1}}{n\pi} + \frac{4}{n^3\pi^3}[(-1)^n - 1] \right) \sin n\pi x.$$

17. Since $f(x)$ is an even function, we expand in a cosine series:

$$a_0 = \frac{2}{\pi} \int_0^\pi (\pi^2 - x^2) \, dx = \frac{4}{3}\pi^2$$

$$a_n = \frac{2}{\pi} \int_0^\pi (\pi^2 - x^2) \cos nx \, dx = \frac{2}{\pi} \left(\frac{\pi^2 - x^2}{n} \sin nx \Big|_0^\pi + \frac{2}{n} \int_0^\pi x \sin nx \, dx \right) = \frac{4}{n^2}(-1)^{n+1}.$$

Thus

$$f(x) = \frac{2}{3}\pi^2 + \sum_{n=1}^{\infty} \frac{4}{n^2}(-1)^{n+1} \cos nx \, dx.$$

18. Since $f(x)$ is an odd function, we expand in a sine series:

$$\begin{aligned} b_n &= \frac{2}{\pi} \int_0^\pi x^3 \sin nx \, dx = \frac{2}{\pi} \left(-\frac{x^3}{n} \cos nx \Big|_0^\pi + \frac{3}{n} \int_0^\pi x^2 \cos nx \, dx \right) = \frac{2\pi^2}{n}(-1)^{n+1} - \frac{12}{n^2\pi} \int_0^\pi x \sin nx \, dx \\ &= \frac{2\pi^2}{n}(-1)^{n+1} - \frac{12}{n^2\pi} \left(-\frac{x}{n} \cos nx \Big|_0^\pi + \frac{1}{n} \int_0^\pi \cos nx \, dx \right) = \frac{2\pi^2}{n}(-1)^{n+1} + \frac{12}{n^3}(-1)^n. \end{aligned}$$

Thus

$$f(x) = \sum_{n=1}^{\infty} \left(\frac{2\pi^2}{n}(-1)^{n+1} + \frac{12}{n^3}(-1)^n \right) \sin nx.$$

19. Since $f(x)$ is an odd function, we expand in a sine series:

$$b_n = \frac{2}{\pi} \int_0^\pi (x+1) \sin nx \, dx = \frac{2(\pi+1)}{n\pi}(-1)^{n+1} + \frac{2}{n\pi}.$$

Thus

$$f(x) = \sum_{n=1}^{\infty} \left(\frac{2(\pi+1)}{n\pi} (-1)^{n+1} + \frac{2}{n\pi} \right) \sin nx.$$

20. Since $f(x)$ is an odd function, we expand in a sine series:

$$\begin{aligned} b_n &= 2 \int_0^1 (x-1) \sin n\pi x \, dx = 2 \left[\int_0^1 x \sin n\pi x \, dx - \int_0^1 \sin n\pi x \, dx \right] \\ &= 2 \left[\frac{1}{n^2\pi^2} \sin n\pi x - \frac{x}{n\pi} \cos n\pi x + \frac{1}{n\pi} \cos n\pi x \right]_0^1 = -\frac{2}{n\pi}. \end{aligned}$$

Thus

$$f(x) = -\sum_{n=1}^{\infty} \frac{2}{n\pi} \sin n\pi x.$$

21. Since $f(x)$ is an even function, we expand in a cosine series:

$$\begin{aligned} a_0 &= \int_0^1 x \, dx + \int_1^2 1 \, dx = \frac{3}{2} \\ a_n &= \int_0^1 x \cos \frac{n\pi}{2} x \, dx + \int_1^2 \cos \frac{n\pi}{2} x \, dx = \frac{4}{n^2\pi^2} \left(\cos \frac{n\pi}{2} - 1 \right). \end{aligned}$$

Thus

$$f(x) = \frac{3}{4} + \sum_{n=1}^{\infty} \frac{4}{n^2\pi^2} \left(\cos \frac{n\pi}{2} - 1 \right) \cos \frac{n\pi}{2} x.$$

22. Since $f(x)$ is an odd function, we expand in a sine series:

$$b_n = \frac{1}{\pi} \int_0^{\pi} x \sin \frac{n}{2} x \, dx + \int_{\pi}^{2\pi} \pi \sin \frac{n}{2} x \, dx = \frac{4}{n^2\pi} \sin \frac{n\pi}{2} + \frac{2}{n} (-1)^{n+1}.$$

Thus

$$f(x) = \sum_{n=1}^{\infty} \left(\frac{4}{n^2\pi} \sin \frac{n\pi}{2} + \frac{2}{n} (-1)^{n+1} \right) \sin \frac{n}{2} x.$$

23. Since $f(x)$ is an even function, we expand in a cosine series:

$$\begin{aligned} a_0 &= \frac{2}{\pi} \int_0^{\pi} \sin x \, dx = \frac{4}{\pi} \\ a_n &= \frac{2}{\pi} \int_0^{\pi} \sin x \cos nx \, dx = \frac{1}{\pi} \int_0^{\pi} \left(\sin(1+n)x + \sin(1-n)x \right) dx \\ &= \frac{2}{\pi(1-n^2)} (1 + (-1)^n) \quad \text{for } n = 2, 3, 4, \dots \\ a_1 &= \frac{1}{\pi} \int_0^{\pi} \sin 2x \, dx = 0. \end{aligned}$$

Thus

$$f(x) = \frac{2}{\pi} + \sum_{n=2}^{\infty} \frac{2[1 + (-1)^n]}{\pi(1-n^2)} \cos nx.$$

12.3 Fourier Cosine and Sine Series

24. Since $f(x)$ is an even function, we expand in a cosine series. [See the solution of Problem 10 in Exercises 12.2 for the computation of the integrals.]

$$a_0 = \frac{2}{\pi/2} \int_0^{\pi/2} \cos x \, dx = \frac{4}{\pi}$$

$$a_n = \frac{2}{\pi/2} \int_0^{\pi/2} \cos x \cos \frac{n\pi}{\pi/2} x \, dx = \frac{4(-1)^{n+1}}{\pi(4n^2 - 1)}$$

Thus

$$f(x) = \frac{2}{\pi} + \sum_{n=1}^{\infty} \frac{4(-1)^{n+1}}{\pi(4n^2 - 1)} \cos 2nx.$$

25. $a_0 = 2 \int_0^{1/2} 1 \, dx = 1$

$$a_n = 2 \int_0^{1/2} 1 \cdot \cos n\pi x \, dx = \frac{2}{n\pi} \sin \frac{n\pi}{2}$$

$$b_n = 2 \int_0^{1/2} 1 \cdot \sin n\pi x \, dx = \frac{2}{n\pi} \left(1 - \cos \frac{n\pi}{2}\right)$$

$$f(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{2}{n\pi} \sin \frac{n\pi}{2} \cos n\pi x$$

$$f(x) = \sum_{n=1}^{\infty} \frac{2}{n\pi} \left(1 - \cos \frac{n\pi}{2}\right) \sin n\pi x$$

26. $a_0 = 2 \int_{1/2}^1 1 \, dx = 1$

$$a_n = 2 \int_{1/2}^1 1 \cdot \cos n\pi x \, dx = -\frac{2}{n\pi} \sin \frac{n\pi}{2}$$

$$b_n = 2 \int_{1/2}^1 1 \cdot \sin n\pi x \, dx = \frac{2}{n\pi} \left(\cos \frac{n\pi}{2} + (-1)^{n+1}\right)$$

$$f(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \left(-\frac{2}{n\pi} \sin \frac{n\pi}{2}\right) \cos n\pi x$$

$$f(x) = \sum_{n=1}^{\infty} \frac{2}{n\pi} \left(\cos \frac{n\pi}{2} + (-1)^{n+1}\right) \sin n\pi x$$

27. $a_0 = \frac{4}{\pi} \int_0^{\pi/2} \cos x \, dx = \frac{4}{\pi}$

$$a_n = \frac{4}{\pi} \int_0^{\pi/2} \cos x \cos 2nx \, dx = \frac{2}{\pi} \int_0^{\pi/2} [\cos(2n+1)x + \cos(2n-1)x] \, dx = \frac{4(-1)^n}{\pi(1-4n^2)}$$

$$b_n = \frac{4}{\pi} \int_0^{\pi/2} \cos x \sin 2nx \, dx = \frac{2}{\pi} \int_0^{\pi/2} [\sin(2n+1)x + \sin(2n-1)x] \, dx = \frac{8n}{\pi(4n^2-1)}$$

$$f(x) = \frac{2}{\pi} + \sum_{n=1}^{\infty} \frac{4(-1)^n}{\pi(1-4n^2)} \cos 2nx$$

$$f(x) = \sum_{n=1}^{\infty} \frac{8n}{\pi(4n^2-1)} \sin 2nx$$

28. $a_0 = \frac{2}{\pi} \int_0^\pi \sin x \, dx = \frac{4}{\pi}$
 $a_n = \frac{2}{\pi} \int_0^\pi \sin x \cos nx \, dx = \frac{1}{\pi} \int_0^\pi [\sin(n+1)x - \sin(n-1)x] \, dx = \frac{2[(-1)^n + 1]}{\pi(1-n^2)} \quad \text{for } n = 2, 3, 4, \dots$
 $b_n = \frac{2}{\pi} \int_0^\pi \sin x \sin nx \, dx = \frac{1}{\pi} \int_0^\pi [\cos(n-1)x - \cos(n+1)x] \, dx = 0 \quad \text{for } n = 2, 3, 4, \dots$
 $a_1 = \frac{1}{\pi} \int_0^\pi \sin 2x \, dx = 0$
 $b_1 = \frac{2}{\pi} \int_0^\pi \sin^2 x \, dx = 1$
 $f(x) = \sin x$
 $f(x) = \frac{2}{\pi} + \frac{2}{\pi} \sum_{n=2}^{\infty} \frac{(-1)^n + 1}{1-n^2} \cos nx$
29. $a_0 = \frac{2}{\pi} \left(\int_0^{\pi/2} x \, dx + \int_{\pi/2}^\pi (\pi - x) \, dx \right) = \frac{\pi}{2}$
 $a_n = \frac{2}{\pi} \left(\int_0^{\pi/2} x \cos nx \, dx + \int_{\pi/2}^\pi (\pi - x) \cos nx \, dx \right) = \frac{2}{n^2\pi} \left(2 \cos \frac{n\pi}{2} + (-1)^{n+1} - 1 \right)$
 $b_n = \frac{2}{\pi} \left(\int_0^{\pi/2} x \sin nx \, dx + \int_{\pi/2}^\pi (\pi - x) \sin nx \, dx \right) = \frac{4}{n^2\pi} \sin \frac{n\pi}{2}$
 $f(x) = \frac{\pi}{4} + \sum_{n=1}^{\infty} \frac{2}{n^2\pi} \left(2 \cos \frac{n\pi}{2} + (-1)^{n+1} - 1 \right) \cos nx$
 $f(x) = \sum_{n=1}^{\infty} \frac{4}{n^2\pi} \sin \frac{n\pi}{2} \sin nx$
30. $a_0 = \frac{1}{\pi} \int_\pi^{2\pi} (x - \pi) \, dx = \frac{\pi}{2}$
 $a_n = \frac{1}{\pi} \int_\pi^{2\pi} (x - \pi) \cos \frac{n}{2}x \, dx = \frac{4}{n^2\pi} \left((-1)^n - \cos \frac{n\pi}{2} \right)$
 $b_n = \frac{1}{\pi} \int_\pi^{2\pi} (x - \pi) \sin \frac{n}{2}x \, dx = \frac{2}{n}(-1)^{n+1} - \frac{4}{n^2\pi} \sin \frac{n\pi}{2}$
 $f(x) = \frac{\pi}{4} + \sum_{n=1}^{\infty} \frac{4}{n^2\pi} \left((-1)^n - \cos \frac{n\pi}{2} \right) \cos \frac{n}{2}x$
 $f(x) = \sum_{n=1}^{\infty} \left(\frac{2}{n}(-1)^{n+1} - \frac{4}{n^2\pi} \sin \frac{n\pi}{2} \right) \sin \frac{n}{2}x$
31. $a_0 = \int_0^1 x \, dx + \int_1^2 1 \, dx = \frac{3}{2}$
 $a_n = \int_0^1 x \cos \frac{n\pi}{2}x \, dx = \frac{4}{n^2\pi^2} \left(\cos \frac{n\pi}{2} - 1 \right)$
 $b_n = \int_0^1 x \sin \frac{n\pi}{2}x \, dx + \int_1^2 1 \cdot \sin \frac{n\pi}{2}x \, dx = \frac{4}{n^2\pi^2} \sin \frac{n\pi}{2} + \frac{2}{n\pi}(-1)^{n+1}$
 $f(x) = \frac{3}{4} + \sum_{n=1}^{\infty} \frac{4}{n^2\pi^2} \left(\cos \frac{n\pi}{2} - 1 \right) \cos \frac{n\pi}{2}x$

12.3 Fourier Cosine and Sine Series

$$f(x) = \sum_{n=1}^{\infty} \left(\frac{4}{n^2\pi^2} \sin \frac{n\pi}{2} + \frac{2}{n\pi} (-1)^{n+1} \right) \sin \frac{n\pi}{2} x$$

$$32. a_0 = \int_0^1 1 dx + \int_1^2 (2-x) dx = \frac{3}{2}$$

$$a_n = \int_0^1 1 \cdot \cos \frac{n\pi}{2} x dx + \int_1^2 (2-x) \cos \frac{n\pi}{2} x dx = \frac{4}{n^2\pi^2} \left(\cos \frac{n\pi}{2} + (-1)^{n+1} \right)$$

$$b_n = \int_0^1 1 \cdot \sin \frac{n\pi}{2} x dx + \int_1^2 (2-x) \sin \frac{n\pi}{2} x dx = \frac{2}{n\pi} + \frac{4}{n^2\pi^2} \sin \frac{n\pi}{2}$$

$$f(x) = \frac{3}{4} + \sum_{n=1}^{\infty} \frac{4}{n^2\pi^2} \left(\cos \frac{n\pi}{2} + (-1)^{n+1} \right) \cos \frac{n\pi}{2} x$$

$$f(x) = \sum_{n=1}^{\infty} \left(\frac{2}{n\pi} + \frac{4}{n^2\pi^2} \sin \frac{n\pi}{2} \right) \sin \frac{n\pi}{2} x$$

$$33. a_0 = 2 \int_0^1 (x^2 + x) dx = \frac{5}{3}$$

$$a_n = 2 \int_0^1 (x^2 + x) \cos n\pi x dx = \frac{2(x^2 + x)}{n\pi} \sin n\pi x \Big|_0^1 - \frac{2}{n\pi} \int_0^1 (2x + 1) \sin n\pi x dx = \frac{2}{n^2\pi^2} [3(-1)^n - 1]$$

$$b_n = 2 \int_0^1 (x^2 + x) \sin n\pi x dx = -\frac{2(x^2 + x)}{n\pi} \cos n\pi x \Big|_0^1 + \frac{2}{n\pi} \int_0^1 (2x + 1) \cos n\pi x dx$$

$$= \frac{4}{n\pi} (-1)^{n+1} + \frac{4}{n^3\pi^3} [(-1)^n - 1]$$

$$f(x) = \frac{5}{6} + \sum_{n=1}^{\infty} \frac{2}{n^2\pi^2} [3(-1)^n - 1] \cos n\pi x$$

$$f(x) = \sum_{n=1}^{\infty} \left(\frac{4}{n\pi} (-1)^{n+1} + \frac{4}{n^3\pi^3} [(-1)^n - 1] \right) \sin n\pi x$$

$$34. a_0 = \int_0^2 (2x - x^2) dx = \frac{4}{3}$$

$$a_n = \int_0^2 (2x - x^2) \cos \frac{n\pi}{2} x dx = \frac{8}{n^2\pi^2} [(-1)^{n+1} - 1]$$

$$b_n = \int_0^2 (2x - x^2) \sin \frac{n\pi}{2} x dx = \frac{16}{n^3\pi^3} [1 - (-1)^n]$$

$$f(x) = \frac{2}{3} + \sum_{n=1}^{\infty} \frac{8}{n^2\pi^2} [(-1)^{n+1} - 1] \cos \frac{n\pi}{2} x$$

$$f(x) = \sum_{n=1}^{\infty} \frac{16}{n^3\pi^3} [1 - (-1)^n] \sin \frac{n\pi}{2} x$$

$$35. a_0 = \frac{1}{\pi} \int_0^{2\pi} x^2 dx = \frac{8}{3} \pi^2$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} x^2 \cos nx dx = \frac{4}{n^2}$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} x^2 \sin nx dx = -\frac{4\pi}{n}$$

$$f(x) = \frac{4}{3}\pi^2 + \sum_{n=1}^{\infty} \left(\frac{4}{n^2} \cos nx - \frac{4\pi}{n} \sin nx \right)$$

$$36. \quad a_0 = \frac{2}{\pi} \int_0^{\pi} x \, dx = \pi$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} x \cos 2nx \, dx = 0$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} x \sin 2nx \, dx = -\frac{1}{n}$$

$$f(x) = \frac{\pi}{2} - \sum_{n=1}^{\infty} \frac{1}{n} \sin 2nx$$

$$37. \quad a_0 = 2 \int_0^1 (x+1) \, dx = 3$$

$$a_n = 2 \int_0^1 (x+1) \cos 2n\pi x \, dx = 0$$

$$b_n = 2 \int_0^1 (x+1) \sin 2n\pi x \, dx = -\frac{1}{n\pi}$$

$$f(x) = \frac{3}{2} - \sum_{n=1}^{\infty} \frac{1}{n\pi} \sin 2n\pi x$$

$$38. \quad a_0 = \frac{2}{2} \int_0^2 (2-x) \, dx = 2$$

$$a_n = \frac{2}{2} \int_0^2 (2-x) \cos n\pi x \, dx = 0$$

$$b_n = \frac{2}{2} \int_0^2 (2-x) \sin n\pi x \, dx = \frac{2}{n\pi}$$

$$f(x) = 1 + \sum_{n=1}^{\infty} \frac{2}{n\pi} \sin n\pi x$$

39. We have

$$b_n = \frac{2}{\pi} \int_0^{\pi} 5 \sin nt \, dt = \frac{10}{n\pi} [1 - (-1)^n]$$

so that

$$f(t) = \sum_{n=1}^{\infty} \frac{10[1 - (-1)^n]}{n\pi} \sin nt.$$

Substituting the assumption $x_p(t) = \sum_{n=1}^{\infty} B_n \sin nt$ into the differential equation then gives

$$x_p'' + 10x_p = \sum_{n=1}^{\infty} B_n(10 - n^2) \sin nt = \sum_{n=1}^{\infty} \frac{10[1 - (-1)^n]}{n\pi} \sin nt$$

and so $B_n = 10[1 - (-1)^n]/n\pi(10 - n^2)$. Thus

$$x_p(t) = \frac{10}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n(10 - n^2)} \sin nt.$$

40. We have

$$b_n = \frac{2}{\pi} \int_0^1 (1-t) \sin n\pi t \, dt = \frac{2}{n\pi}$$

12.3 Fourier Cosine and Sine Series

so that

$$f(t) = \sum_{n=1}^{\infty} \frac{2}{n\pi} \sin n\pi t.$$

Substituting the assumption $x_p(t) = \sum_{n=1}^{\infty} B_n \sin n\pi t$ into the differential equation then gives

$$x_p'' + 10x_p = \sum_{n=1}^{\infty} B_n(10 - n^2\pi^2) \sin n\pi t = \sum_{n=1}^{\infty} \frac{2}{n\pi} \sin n\pi t$$

and so $B_n = 2/n\pi(10 - n^2\pi^2)$. Thus

$$x_p(t) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n(10 - n^2\pi^2)} \sin n\pi t.$$

41. We have

$$a_0 = \frac{2}{\pi} \int_0^{\pi} (2\pi t - t^2) dt = \frac{4}{3}\pi^2$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} (2\pi t - t^2) \cos nt dt = -\frac{4}{n^2}$$

so that

$$f(t) = \frac{2\pi^2}{3} - \sum_{n=1}^{\infty} \frac{4}{n^2} \cos nt.$$

Substituting the assumption

$$x_p(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos nt$$

into the differential equation then gives

$$\frac{1}{4} x_p'' + 12x_p = 6A_0 + \sum_{n=1}^{\infty} A_n \left(-\frac{1}{4}n^2 + 12 \right) \cos nt = \frac{2\pi^2}{3} - \sum_{n=1}^{\infty} \frac{4}{n^2} \cos nt$$

and $A_0 = \pi^2/9$, $A_n = 16/n^2(n^2 - 48)$. Thus

$$x_p(t) = \frac{\pi^2}{18} + 16 \sum_{n=1}^{\infty} \frac{1}{n^2(n^2 - 48)} \cos nt.$$

42. We have

$$a_0 = \frac{2}{1/2} \int_0^{1/2} t dt = \frac{1}{2}$$

$$a_n = \frac{2}{1/2} \int_0^{1/2} t \cos 2n\pi t dt = \frac{1}{n^2\pi^2} [(-1)^n - 1]$$

so that

$$f(t) = \frac{1}{4} + \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2\pi^2} \cos 2n\pi t.$$

Substituting the assumption

$$x_p(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos 2n\pi t$$

into the differential equation then gives

$$\frac{1}{4} x_p'' + 12x_p = 6A_0 + \sum_{n=1}^{\infty} A_n(12 - n^2\pi^2) \cos 2n\pi t = \frac{1}{4} + \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2\pi^2} \cos 2n\pi t$$

and $A_0 = 1/24$, $A_n = [(-1)^n - 1]/n^2\pi^2(12 - n^2\pi^2)$. Thus

$$x_p(t) = \frac{1}{48} + \frac{1}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2(12 - n^2\pi^2)} \cos 2n\pi t.$$

43. (a) The general solution is $x(t) = c_1 \cos \sqrt{10}t + c_2 \sin \sqrt{10}t + x_p(t)$, where

$$x_p(t) = \frac{10}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n(10 - n^2)} \sin nt.$$

The initial condition $x(0) = 0$ implies $c_1 + x_p(0) = 0$. Since $x_p(0) = 0$, we have $c_1 = 0$ and $x(t) = c_2 \sin \sqrt{10}t + x_p(t)$. Then $x'(t) = c_2 \sqrt{10} \cos \sqrt{10}t + x'_p(t)$ and $x'(0) = 0$ implies

$$c_2 \sqrt{10} + \frac{10}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{10 - n^2} \cos 0 = 0.$$

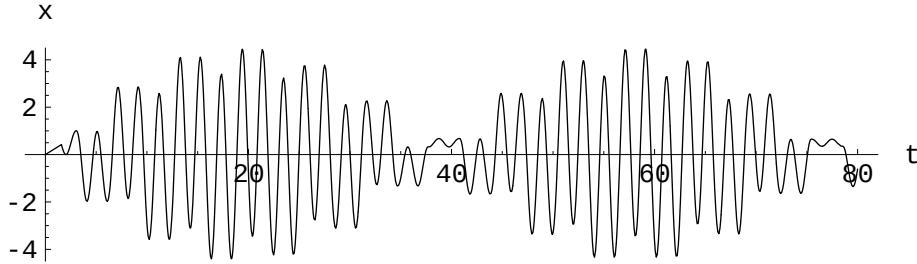
Thus

$$c_2 = -\frac{\sqrt{10}}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{10 - n^2}$$

and

$$x(t) = \frac{10}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{10 - n^2} \left[\frac{1}{n} \sin nt - \frac{1}{\sqrt{10}} \sin \sqrt{10}t \right].$$

- (b) The graph is plotted using eight nonzero terms in the series expansion of $x(t)$.



44. (a) The general solution is $x(t) = c_1 \cos 4\sqrt{3}t + c_2 \sin 4\sqrt{3}t + x_p(t)$, where

$$x_p(t) = \frac{\pi^2}{18} + 16 \sum_{n=1}^{\infty} \frac{1}{n^2(n^2 - 48)} \cos nt.$$

The initial condition $x(0) = 0$ implies $c_1 + x_p(0) = 1$ or

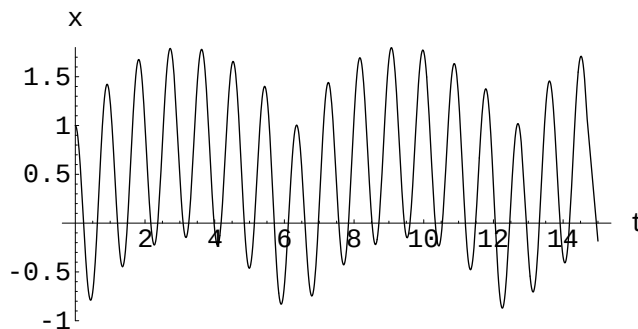
$$c_1 = 1 - x_p(0) = 1 - \frac{\pi^2}{18} - 16 \sum_{n=1}^{\infty} \frac{1}{n^2(n^2 - 48)}.$$

Now $x'(t) = -4\sqrt{3}c_1 \sin 4\sqrt{3}t + 4\sqrt{3}c_2 \cos 4\sqrt{3}t + x'_p(t)$, so $x'(0) = 0$ implies $4\sqrt{3}c_2 + x'_p(0) = 0$. Since $x'_p(0) = 0$, we have $c_2 = 0$ and

$$\begin{aligned} x(t) &= \left(1 - \frac{\pi^2}{18} - 16 \sum_{n=1}^{\infty} \frac{1}{n^2(n^2 - 48)} \right) \cos 4\sqrt{3}t + \frac{\pi^2}{18} + 16 \sum_{n=1}^{\infty} \frac{1}{n^2(n^2 - 48)} \cos nt \\ &= \frac{\pi^2}{18} + \left(1 - \frac{\pi^2}{18} \right) \cos 4\sqrt{3}t + 16 \sum_{n=1}^{\infty} \frac{1}{n^2(n^2 - 48)} [\cos nt - \cos 4\sqrt{3}t]. \end{aligned}$$

12.3 Fourier Cosine and Sine Series

(b) The graph is plotted using five nonzero terms in the series expansion of $x(t)$.



45. (a) We have

$$b_n = \frac{2}{L} \int_0^L \frac{w_0 x}{L} \sin \frac{n\pi}{L} x dx = \frac{2w_0}{n\pi} (-1)^{n+1}$$

so that

$$w(x) = \sum_{n=1}^{\infty} \frac{2w_0}{n\pi} (-1)^{n+1} \sin \frac{n\pi}{L} x.$$

(b) If we assume $y_p(x) = \sum_{n=1}^{\infty} B_n \sin(n\pi x/L)$ then

$$y_p^{(4)} = \sum_{n=1}^{\infty} \frac{n^4 \pi^4}{L^4} B_n \sin \frac{n\pi}{L} x$$

and so the differential equation $EI y_p^{(4)} = w(x)$ gives

$$B_n = \frac{2w_0(-1)^{n+1} L^4}{EI n^5 \pi^5}.$$

Thus

$$y_p(x) = \frac{2w_0 L^4}{EI \pi^5} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^5} \sin \frac{n\pi}{L} x.$$

46. We have

$$b_n = \frac{2}{L} \int_{L/3}^{2L/3} w_0 \sin \frac{n\pi}{L} x dx = \frac{2w_0}{n\pi} \left(\cos \frac{n\pi}{3} - \cos \frac{2n\pi}{3} \right)$$

so that

$$w(x) = \sum_{n=1}^{\infty} \frac{2w_0}{n\pi} \left(\cos \frac{n\pi}{3} - \cos \frac{2n\pi}{3} \right) \sin \frac{n\pi}{L} x.$$

If we assume $y_p(x) = \sum_{n=1}^{\infty} B_n \sin(n\pi x/L)$ then

$$y_p^{(4)}(x) = \sum_{n=1}^{\infty} \frac{n^4 \pi^4}{L^4} B_n \sin \frac{n\pi}{L} x$$

and so the differential equation $EI y_p^{(4)}(x) = w(x)$ gives

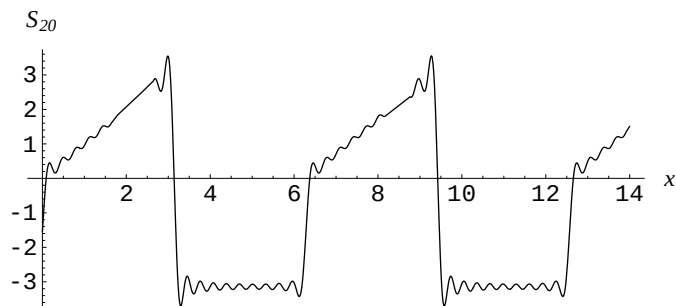
$$B_n = 2w_0 L^4 \frac{\cos \frac{n\pi}{3} - \cos \frac{2n\pi}{3}}{EI n^5 \pi^5}.$$

Thus

$$y_p(x) = \frac{2w_0 L^4}{EI \pi^5} \sum_{n=1}^{\infty} \frac{\cos \frac{n\pi}{3} - \cos \frac{2n\pi}{3}}{n^5} \sin \frac{n\pi}{L} x.$$

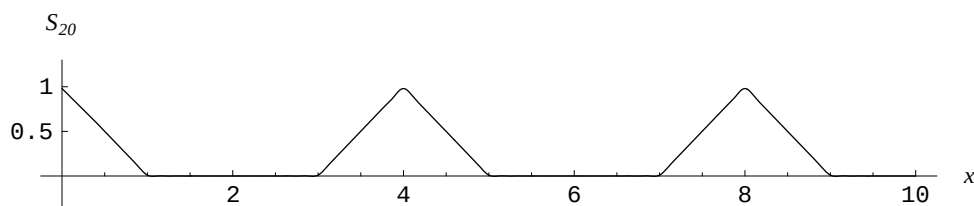
47. The graph is obtained by summing the series from $n = 1$ to 20. It appears that

$$f(x) = \begin{cases} x, & 0 < x < \pi \\ -\pi, & \pi < x < 2\pi. \end{cases}$$



48. The graph is obtained by summing the series from $n = 1$ to 10. It appears that

$$f(x) = \begin{cases} 1 - x, & 0 < x < 1 \\ 0, & 1 < x < 2. \end{cases}$$



49. The function in Problem 47 is not unique; it could also be defined as

$$f(x) = \begin{cases} x, & 0 < x < \pi \\ 1, & x = \pi \\ -\pi, & \pi < x < 2\pi. \end{cases}$$

The function in Problem 48 is not unique; it could also be defined as

$$f(x) = \begin{cases} 0, & -2 < x < -1 \\ x + 1, & -1 < x < 0 \\ -x + 1, & 0 < x < 1 \\ 0, & 1 < x < 2. \end{cases}$$

50. The cosine series converges to an even extension of the function on the interval $(-\pi, 0)$. Since the even extension of $f(x)$ is $f(-x)$, in this case $f(-x) = e^{-x}$ on $(-\pi, 0)$.
51. No, it is not a full Fourier series. A full Fourier series of $f(x) = e^x, 0 < x < \pi$, would converge to the π -periodic extension of f . The cosine and sine series converge to a 2π -periodic extension (even and odd, respectively). The average of the two series converges to a 2π -periodic extension of

$$f(x) = \begin{cases} e^x, & 0 < x < \pi \\ 0, & -\pi < x < 0. \end{cases}$$

52. (a) If f and g are even and $h(x) = f(x)g(x)$ then

$$h(-x) = f(-x)g(-x) = f(x)g(x) = h(x)$$

and h is even.

12.3 Fourier Cosine and Sine Series

(c) If f is even and g is odd and $h(x) = f(x)g(x)$ then

$$h(-x) = f(-x)g(-x) = f(x)[-g(x)] = -h(x)$$

and h is odd.

(d) Let $h(x) = f(x) \pm g(x)$ where f and g are even. Then

$$h(-x) = f(-x) \pm g(-x) = f(x) \pm g(x) = h(x),$$

and so h is an even function.

(f) If f is even then

$$\int_{-a}^a f(x) dx = -\int_a^0 f(-u) du + \int_0^a f(x) dx = \int_0^a f(u) du + \int_0^a f(x) dx = 2 \int_0^a f(x) dx.$$

(g) If f is odd then

$$\begin{aligned} \int_{-a}^a f(x) dx &= -\int_{-a}^0 f(-x) dx + \int_0^a f(x) dx = \int_a^0 f(u) du + \int_0^a f(x) dx \\ &= -\int_0^a f(u) du + \int_0^a f(x) dx = 0. \end{aligned}$$

EXERCISES 12.4

Complex Fourier Series

In this section we make use of the following identities due to Euler's formula:

$$e^{in\pi} = e^{-in\pi} = (-1)^n, \quad e^{-2in\pi} = 1, \quad e^{-in\pi/2} = (-i)^n.$$

1. Identifying $p = 2$ we have

$$\begin{aligned} c_n &= \frac{1}{4} \int_{-2}^2 f(x) e^{-in\pi x/2} dx = \frac{1}{4} \left[\int_{-2}^0 (-1) e^{-in\pi x/2} dx + \int_0^2 e^{-in\pi x/2} dx \right] \\ &= \frac{i}{2n\pi} [-1 + e^{in\pi} + e^{-in\pi} - 1] = \frac{i}{2n\pi} [-1 + (-1)^n + (-1)^n - 1] = \frac{1 - (-1)^n}{n\pi i} \end{aligned}$$

and

$$c_0 = \frac{1}{4} \int_{-2}^2 f(x) dx = 0.$$

Thus

$$f(x) = \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{1 - (-1)^n}{in\pi} e^{in\pi x/2}.$$

2. Identifying $2p = 2$ or $p = 1$ we have

$$\begin{aligned} c_n &= \frac{1}{2} \int_0^2 f(x) e^{-in\pi x} dx = \frac{1}{2} \int_1^2 e^{-in\pi x} dx = -\frac{1}{2in\pi} e^{-in\pi x} \Big|_1^2 \\ &= -\frac{1}{2in\pi} (e^{-2in\pi} - e^{-in\pi}) = -\frac{1}{2in\pi} [1 - (-1)^n] = \frac{i}{2n\pi} [1 - (-1)^n] \end{aligned}$$

and

$$c_0 = \frac{1}{2} \int_0^2 f(x) dx = \frac{1}{2} \int_1^2 dx = \frac{1}{2}.$$

Thus

$$f(x) = \frac{1}{2} + \frac{i}{2\pi} \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{1 - (-1)^n}{n} e^{in\pi x}.$$

3. Identifying $p = 1/2$ we have

$$\begin{aligned} c_n &= \int_{-1/2}^{1/2} f(x) e^{-2in\pi x} dx = \int_0^{1/4} e^{-2in\pi x} dx = -\frac{1}{2in\pi} e^{-2in\pi x} \Big|_0^{1/4} \\ &= -\frac{1}{2in\pi} [e^{-in\pi/2} - 1] = -\frac{1}{2in\pi} [(-i)^n - 1] = \frac{i}{2n\pi} [(-i)^n - 1] \end{aligned}$$

and

$$c_0 = \int_0^{1/4} dx = \frac{1}{4}.$$

Thus

$$f(x) = \frac{1}{4} + \frac{i}{2\pi} \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{(-i)^n - 1}{n} e^{2in\pi x}.$$

4. Identifying $p = \pi$ we have

$$\begin{aligned} c_n &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx/\pi} dx = \frac{1}{2\pi} \int_0^{\pi} x e^{-inx/\pi} dx \\ &= \frac{1}{2} \left(\frac{\pi}{n^2} + \frac{ix}{n} \right) e^{-inx/\pi} \Big|_0^{\pi} = \frac{\pi(1+in)}{2n^2} e^{-in} - \frac{\pi}{2n^2} \end{aligned}$$

and

$$c_0 = \frac{1}{2\pi} \int_0^{\pi} x dx = \frac{\pi}{4}.$$

Thus

$$f(x) = \frac{\pi}{4} + \frac{\pi}{2} \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{1}{n^2} [(1+in)e^{-in} - 1] e^{inx}.$$

5. Identifying $2p = 2\pi$ or $p = \pi$ we have

$$\begin{aligned} c_n &= \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{-inx} dx = \frac{1}{2\pi} \int_0^{2\pi} x e^{-inx} dx \\ &= \frac{1}{2\pi} \left(\frac{1}{n^2} + \frac{ix}{n} \right) e^{-inx} \Big|_0^{2\pi} = \frac{1+2in\pi}{2n^2\pi} - \frac{1}{2n^2\pi} = \frac{i}{n} \end{aligned}$$

12.4 Complex Fourier Series

and

$$c_0 = \frac{1}{2\pi} \int_0^{2\pi} x \, dx = \pi.$$

Thus

$$f(x) = \pi + \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{i}{n} e^{inx}.$$

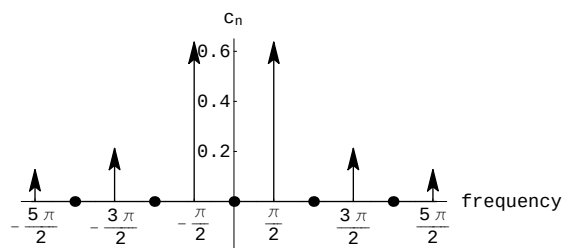
6. Identifying $p = 1$ we have

$$\begin{aligned} c_n &= \frac{1}{2} \int_{-1}^1 f(x) e^{-in\pi x} dx = \frac{1}{2} \left[\int_{-1}^0 e^x e^{-in\pi x} dx + \int_0^1 e^{-x} e^{-in\pi x} dx \right] \\ &= \frac{1}{2} \left[-\frac{1}{1-in\pi} e^{(1-in\pi)x} \Big|_{-1}^0 - \frac{1}{1+in\pi} e^{-(1+in\pi)x} \Big|_0^1 \right] \\ &= \frac{e - (-1)^n}{e(1-in\pi)} + \frac{1 - e^{-1}(-1)^n}{1+in\pi} = \frac{2[e - (-1)^n]}{e(1+n^2\pi^2)}. \end{aligned}$$

Thus

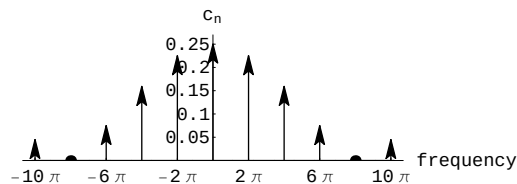
$$f(x) = \sum_{n=-\infty}^{\infty} \frac{2[e - (-1)^n]}{e(1+n^2\pi^2)} e^{in\pi x}.$$

7. The fundamental period is $T = 4$, so $\omega = 2\pi/4 = \pi/2$ and the values of $n\omega$ are $0, \pm\pi/2, \pm\pi, \pm3\pi/2, \dots$. From Problem 1, $c_0 = 0$ and $|c_n| = (1 - (-1)^n)/n\pi$. The table shows some values of n with corresponding values of $|c_n|$. The graph is a portion of the frequency spectrum.



n	-5	-4	-3	-2	-1	0	1	2	3	4	5
c_n	0.1273	0.0000	0.2122	0.0000	0.6366	0.0000	0.6366	0.0000	0.2122	0.0000	0.1273

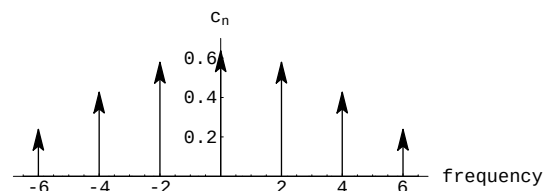
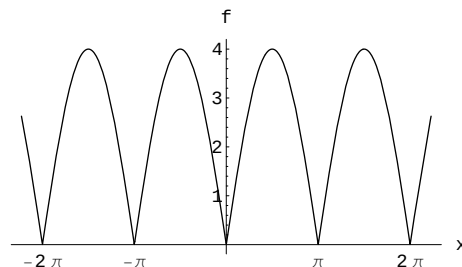
8. The fundamental period is $T = 1$, so $\omega = 2\pi$ and the values of $n\omega$ are $0, \pm2\pi, \pm4\pi, \pm6\pi, \dots$. From Problem 3, $c_0 = \frac{1}{4}$ and $|c_n| = |(-i)^n - 1|/2n\pi$, or $c_1 = c_{-1} = \sqrt{2}/2\pi$, $c_2 = c_{-2} = 1/2\pi$, $c_3 = c_{-3} = \sqrt{2}/6\pi$, $c_4 = c_{-4} = 0$, $c_5 = c_{-5} = \sqrt{2}/10\pi$, $c_6 = c_{-6} = 1/6\pi$, $c_7 = c_{-7} = \sqrt{2}/14\pi$, $c_8 = c_{-8} = 0, \dots$. The table shows some values of n with corresponding values of $|c_n|$. The graph is a portion of the frequency spectrum.



n	-5	-4	-3	-2	-1	0	1	2	3	4	5
c_n	0.0450	0.0000	0.0750	0.1592	0.2251	0.2500	0.2251	0.1592	0.0750	0.0000	0.0450

9. Identifying $2p = \pi$ or $p = \pi/2$, and using $\sin x = (e^{ix} - e^{-ix})/2i$, we have

$$\begin{aligned} c_n &= \frac{1}{\pi} \int_0^\pi f(x) e^{-2inx/\pi} dx = \frac{1}{\pi} \int_0^\pi (\sin x) e^{-2inx/\pi} dx \\ &= \frac{1}{\pi} \int_0^\pi \frac{1}{2i} (e^{ix} - e^{-ix}) e^{-2inx/\pi} dx \\ &= \frac{1}{2\pi i} \int_0^\pi \left(e^{(1-2n/\pi)ix} - e^{-(1+2n/\pi)ix} \right) dx \\ &= \frac{1}{2\pi i} \left[\frac{1}{i(1-2n/\pi)} e^{(1-2n/\pi)ix} \right. \\ &\quad \left. + \frac{1}{i(1+2n/\pi)} e^{-(1+2n/\pi)ix} \right]_0^\pi \\ &= \frac{\pi(1 + e^{-2in})}{\pi^2 - 4n^2}. \end{aligned}$$

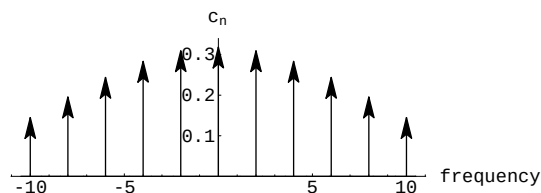
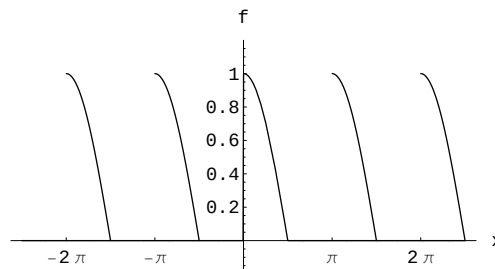


The fundamental period is $T = \pi$, so $\omega = 2\pi/\pi = 2$ and the values of $n\omega$ are $0, \pm 2, \pm 4, \pm 6, \dots$. Values of $|c_n|$ for $n = 0, \pm 1, \pm 2, \pm 3, \pm 4$, and ± 5 are shown in the table. The bottom graph is a portion of the frequency spectrum.

n	-5	-4	-3	-2	-1	0	1	2	3	4	5
c_n	0.0198	0.0759	0.2380	0.4265	0.5784	0.6366	0.5784	0.4265	0.2380	0.0759	0.0198

10. Identifying $2p = \pi$ or $p = \pi/2$, and using $\cos x = (e^{ix} + e^{-ix})/2$, we have

$$\begin{aligned} c_n &= \frac{1}{\pi} \int_0^\pi f(x) e^{-2inx/\pi} dx \\ &= \frac{1}{\pi} \int_0^{\pi/2} (\cos x) e^{-2inx/\pi} dx \\ &= \frac{1}{\pi} \int_0^{\pi/2} \frac{1}{2} (e^{ix} + e^{-ix}) e^{-2inx/\pi} dx \\ &= \frac{1}{2\pi} \int_0^{\pi/2} \left(e^{(1-2n/\pi)ix} + e^{-(1+2n/\pi)ix} \right) dx \\ &= \frac{1}{2\pi} \left[\frac{1}{i(1-2n/\pi)} e^{(1-2n/\pi)ix} \right. \\ &\quad \left. + \frac{1}{i(1+2n/\pi)} e^{-(1+2n/\pi)ix} \right]_0^{\pi/2} \\ &= \frac{2ne^{-in} + i\pi}{\pi^2 - 4n^2}. \end{aligned}$$



The fundamental period is $T = \pi$, so $\omega = 2\pi/\pi = 2$ and the values of $n\omega$ are $0, \pm 2, \pm 4, \pm 6, \dots$. Values of $|c_n|$ for $n = 0, \pm 1, \pm 2, \pm 3, \pm 4$, and ± 5 are shown in the table. The bottom graph is a portion of the frequency spectrum.

n	-5	-4	-3	-2	-1	0	1	2	3	4	5
c_n	0.1447	0.1954	0.2437	0.2833	0.3093	0.3183	0.3093	0.2833	0.2437	0.1954	0.1447

12.4 Complex Fourier Series

11. (a) Adding $c_n = \frac{1}{2}(a_n - ib_n)$ and $c_{-n} = \frac{1}{2}(a_n + ib_n)$ we get $c_n + c_{-n} = a_n$. Subtracting, we get $c_n - c_{-n} = -ib_n$. Multiplying both sides by i we obtain $i(c_n - c_{-n}) = b_n$.

(b) From

$$a_n = c_n + c_{-n} = (-1)^n \frac{\sinh \pi}{\pi} \left[\frac{1 - in}{n^2 + 1} + \frac{1 + in}{n^2 + 1} \right] = \frac{2(-1)^n \sinh \pi}{\pi(n^2 + 1)}, \quad n = 0, 1, 2, \dots$$

and

$$b_n = i(c_n - c_{-n}) = i(-1)^n \frac{\sinh \pi}{\pi} \left[\frac{1 - in}{n^2 + 1} - \frac{1 + in}{n^2 + 1} \right] = i(-1)^n \frac{\sinh \pi}{\pi} \left[-\frac{2in}{n^2 + 1} \right] = \frac{2(-1)^n n \sinh \pi}{\pi(n^2 + 1)},$$

the Fourier series of f is

$$f(x) = \frac{\sinh \pi}{\pi} + \frac{2 \sinh \pi}{\pi} \sum_{n=1}^{\infty} \left[\frac{(-1)^n}{n^2 + 1} \cos nx + \frac{n(-1)^n}{n^2 + 1} \sin nx \right].$$

12. From Problem 11 and the fact that f is odd, $c_n + c_{-n} = a_n = 0$, so $c_{-n} = -c_n$. Then $b_n = i(c_n - c_{-n}) = 2ic_n$. From Problem 1, $b_n = 2i[1 - (-1)^n]/n\pi = 2[1 - (-1)^n]/n\pi$, and the Fourier sine series of f is

$$f(x) = \sum_{n=1}^{\infty} \frac{2[1 - (-1)^n]}{n\pi} \sin \frac{n\pi x}{2}.$$

EXERCISES 12.5

Sturm-Liouville Problem

1. For $\lambda \leq 0$ the only solution of the boundary-value problem is $y = 0$. For $\lambda = \alpha^2 > 0$ we have

$$y = c_1 \cos \alpha x + c_2 \sin \alpha x.$$

Now

$$y'(x) = -c_1 \alpha \sin \alpha x + c_2 \alpha \cos \alpha x$$

and $y'(0) = 0$ implies $c_2 = 0$, so

$$y(1) + y'(1) = c_1(\cos \alpha - \alpha \sin \alpha) = 0 \quad \text{or} \quad \cot \alpha = \alpha.$$

The eigenvalues are $\lambda_n = \alpha_n^2$ where $\alpha_1, \alpha_2, \alpha_3, \dots$ are the consecutive positive solutions of $\cot \alpha = \alpha$. The corresponding eigenfunctions are $\cos \alpha_n x$ for $n = 1, 2, 3, \dots$. Using a CAS we find that the first four eigenvalues are approximately 0.7402, 11.7349, 41.4388, and 90.8082 with corresponding approximate eigenfunctions $\cos 0.8603x$, $\cos 3.4256x$, $\cos 6.4373x$, and $\cos 9.5293x$.

2. For $\lambda < 0$ the only solution of the boundary-value problem is $y = 0$. For $\lambda = 0$ we have $y = c_1 x + c_2$. Now $y' = c_1$ and the boundary conditions both imply $c_1 + c_2 = 0$. Thus, $\lambda = 0$ is an eigenvalue with corresponding eigenfunction $y_0 = x - 1$.

For $\lambda = \alpha^2 > 0$ we have

$$y = c_1 \cos \alpha x + c_2 \sin \alpha x$$

and

$$y'(x) = -c_1\alpha \sin \alpha x + c_2\alpha \cos \alpha x.$$

The boundary conditions imply

$$\begin{aligned} c_1 + c_2\alpha &= 0 \\ c_1 \cos \alpha + c_2 \sin \alpha &= 0 \end{aligned}$$

which gives

$$-c_2\alpha \cos \alpha + c_2 \sin \alpha = 0 \quad \text{or} \quad \tan \alpha = \alpha.$$

The eigenvalues are $\lambda_n = \alpha_n^2$ where $\alpha_1, \alpha_2, \alpha_3, \dots$ are the consecutive positive solutions of $\tan \alpha = \alpha$. The corresponding eigenfunctions are $\alpha \cos \alpha x - \sin \alpha x$ (obtained by taking $c_2 = -1$ in the first equation of the system.) Using a CAS we find that the first four positive eigenvalues are 20.1907, 59.6795, 118.9000, and 197.858 with corresponding eigenfunctions $4.4934 \cos 4.4934x - \sin 4.4934x$, $7.7253 \cos 7.7253x - \sin 7.7253x$, $10.9041 \cos 10.9041x - \sin 10.9041x$, and $14.0662 \cos 14.0662x - \sin 14.0662x$.

3. For $\lambda = 0$ the solution of $y'' = 0$ is $y = c_1x + c_2$. The condition $y'(0) = 0$ implies $c_1 = 0$, so $\lambda = 0$ is an eigenvalue with corresponding eigenfunction 1.

For $\lambda = -\alpha^2 < 0$ we have $y = c_1 \cosh \alpha x + c_2 \sinh \alpha x$ and $y' = c_1\alpha \sinh \alpha x + c_2\alpha \cosh \alpha x$. The condition $y'(0) = 0$ implies $c_2 = 0$ and so $y = c_1 \cosh \alpha x$. Now the condition $y'(L) = 0$ implies $c_1 = 0$. Thus $y = 0$ and there are no negative eigenvalues.

For $\lambda = \alpha^2 > 0$ we have $y = c_1 \cos \alpha x + c_2 \sin \alpha x$ and $y' = -c_1\alpha \sin \alpha x + c_2\alpha \cos \alpha x$. The condition $y'(0) = 0$ implies $c_2 = 0$ and so $y = c_1 \cos \alpha x$. Now the condition $y'(L) = 0$ implies $-c_1\alpha \sin \alpha L = 0$. For $c_1 \neq 0$ this condition will hold when $\alpha L = n\pi$ or $\lambda = \alpha^2 = n^2\pi^2/L^2$, where $n = 1, 2, 3, \dots$. These are the positive eigenvalues with corresponding eigenfunctions $\cos(n\pi x/L)$, $n = 1, 2, 3, \dots$.

4. For $\lambda = -\alpha^2 < 0$ we have

$$\begin{aligned} y &= c_1 \cosh \alpha x + c_2 \sinh \alpha x \\ y' &= c_1\alpha \sinh \alpha x + c_2\alpha \cosh \alpha x. \end{aligned}$$

Using the fact that $\cosh x$ is an even function and $\sinh x$ is odd we have

$$\begin{aligned} y(-L) &= c_1 \cosh(-\alpha L) + c_2 \sinh(-\alpha L) \\ &= c_1 \cosh \alpha L - c_2 \sinh \alpha L \end{aligned}$$

and

$$\begin{aligned} y'(-L) &= c_1\alpha \sinh(-\alpha L) + c_2\alpha \cosh(-\alpha L) \\ &= -c_1\alpha \sinh \alpha L + c_2\alpha \cosh \alpha L. \end{aligned}$$

The boundary conditions imply

$$c_1 \cosh \alpha L - c_2 \sinh \alpha L = c_1 \cosh \alpha L + c_2 \sinh \alpha L$$

or

$$2c_2 \sinh \alpha L = 0$$

and

$$-c_1\alpha \sinh \alpha L + c_2\alpha \cosh \alpha L = c_1\alpha \sinh \alpha L + c_2\alpha \cosh \alpha L$$

or

$$2c_1\alpha \sinh \alpha L = 0.$$

12.5 Sturm-Liouville Problem

Since $\alpha L \neq 0$, $c_1 = c_2 = 0$ and the only solution of the boundary-value problem in this case is $y = 0$.

For $\lambda = 0$ we have

$$\begin{aligned}y &= c_1 x + c_2 \\y' &= c_1.\end{aligned}$$

From $y(-L) = y(L)$ we obtain

$$-c_1 L + c_2 = c_1 L + c_2.$$

Then $c_1 = 0$ and $y = 1$ is an eigenfunction corresponding to the eigenvalue $\lambda = 0$.

For $\lambda = \alpha^2 > 0$ we have

$$\begin{aligned}y &= c_1 \cos \alpha x + c_2 \sin \alpha x \\y' &= -c_1 \alpha \sin \alpha x + c_2 \alpha \cos \alpha x.\end{aligned}$$

The first boundary condition implies

$$c_1 \cos \alpha L - c_2 \sin \alpha L = c_1 \cos \alpha L + c_2 \sin \alpha L$$

or

$$2c_2 \sin \alpha L = 0.$$

Thus, if $c_1 = 0$ and $c_2 \neq 0$,

$$\alpha L = n\pi \quad \text{or} \quad \lambda = \alpha^2 = \frac{n^2 \pi^2}{L^2}, \quad n = 1, 2, 3, \dots$$

The corresponding eigenfunctions are $\sin(n\pi x/L)$, for $n = 1, 2, 3, \dots$. Similarly, the second boundary condition implies

$$2c_1 \alpha \sin \alpha L = 0.$$

If $c_1 \neq 0$ and $c_2 = 0$,

$$\alpha L = n\pi \quad \text{or} \quad \lambda = \alpha^2 = \frac{n^2 \pi^2}{L^2}, \quad n = 1, 2, 3, \dots,$$

and the corresponding eigenfunctions are $\cos(n\pi x/L)$, for $n = 1, 2, 3, \dots$.

5. The eigenfunctions are $\cos \alpha_n x$ where $\cot \alpha_n = \alpha_n$. Thus

$$\begin{aligned}\|\cos \alpha_n x\|^2 &= \int_0^1 \cos^2 \alpha_n x \, dx = \frac{1}{2} \int_0^1 (1 + \cos 2\alpha_n x) \, dx \\&= \frac{1}{2} \left(x + \frac{1}{2\alpha_n} \sin 2\alpha_n x \right) \Big|_0^1 = \frac{1}{2} \left(1 + \frac{1}{2\alpha_n} \sin 2\alpha_n \right) \\&= \frac{1}{2} \left[1 + \frac{1}{2\alpha_n} (2 \sin \alpha_n \cos \alpha_n) \right] \\&= \frac{1}{2} \left[1 + \frac{1}{\alpha_n} \sin \alpha_n \cot \alpha_n \sin \alpha_n \right] \\&= \frac{1}{2} \left[1 + \frac{1}{\alpha_n} (\sin \alpha_n) \alpha_n (\sin \alpha_n) \right] = \frac{1}{2} (1 + \sin^2 \alpha_n).\end{aligned}$$

6. The eigenfunctions are $\sin \alpha_n x$ where $\tan \alpha_n = -\alpha_n$. Thus

$$\begin{aligned}\|\sin \alpha_n x\|^2 &= \int_0^1 \sin^2 \alpha_n x \, dx = \frac{1}{2} \int_0^1 (1 - \cos 2\alpha_n x) \, dx \\ &= \frac{1}{2} \left(x - \frac{1}{2\alpha_n} \sin 2\alpha_n x \right) \Big|_0^1 = \frac{1}{2} \left(1 - \frac{1}{2\alpha_n} \sin 2\alpha_n \right) \\ &= \frac{1}{2} \left[1 - \frac{1}{2\alpha_n} (2 \sin \alpha_n \cos \alpha_n) \right] \\ &= \frac{1}{2} \left[1 - \frac{1}{\alpha_n} \tan \alpha_n \cos \alpha_n \cos \alpha_n \right] \\ &= \frac{1}{2} \left[1 - \frac{1}{\alpha_n} (-\alpha_n \cos^2 \alpha_n) \right] = \frac{1}{2} (1 + \cos^2 \alpha_n).\end{aligned}$$

7. (a) If $\lambda \leq 0$ the initial conditions imply $y = 0$. For $\lambda = \alpha^2 > 0$ the general solution of the Cauchy-Euler differential equation is $y = c_1 \cos(\alpha \ln x) + c_2 \sin(\alpha \ln x)$. The condition $y(1) = 0$ implies $c_1 = 0$, so that $y = c_2 \sin(\alpha \ln x)$. The condition $y(5) = 0$ implies $\alpha \ln 5 = n\pi$, $n = 1, 2, 3, \dots$. Thus, the eigenvalues are $n^2\pi^2/(\ln 5)^2$ for $n = 1, 2, 3, \dots$, with corresponding eigenfunctions $\sin[(n\pi/\ln 5) \ln x]$.

- (b) The self-adjoint form is

$$\frac{d}{dx}[xy'] + \frac{\lambda}{x}y = 0.$$

- (c) An orthogonality relation is

$$\int_1^5 \frac{1}{x} \sin\left(\frac{m\pi}{\ln 5} \ln x\right) \sin\left(\frac{n\pi}{\ln 5} \ln x\right) dx = 0, \quad m \neq n.$$

8. (a) The roots of the auxiliary equation $m^2 + m + \lambda = 0$ are $\frac{1}{2}(-1 \pm \sqrt{1-4\lambda})$. When $\lambda = 0$ the general solution of the differential equation is $c_1 + c_2 e^{-x}$. The boundary conditions imply $c_1 + c_2 = 0$ and $c_1 + c_2 e^{-2} = 0$. Since the determinant of the coefficients is not 0, the only solution of this homogeneous system is $c_1 = c_2 = 0$, in which case $y = 0$. When $\lambda = \frac{1}{4}$, the general solution of the differential equation is $c_1 e^{-x/2} + c_2 x e^{-x/2}$. The boundary conditions imply $c_1 = 0$ and $c_1 + 2c_2 = 0$, so $c_1 = c_2 = 0$ and $y = 0$. Similarly, if $0 < \lambda < \frac{1}{4}$, the general solution is

$$y = c_1 e^{\frac{1}{2}(-1+\sqrt{1-4\lambda})x} + c_2 e^{\frac{1}{2}(-1-\sqrt{1-4\lambda})x}.$$

In this case the boundary conditions again imply $c_1 = c_2 = 0$, and so $y = 0$. Now, for $\lambda > \frac{1}{4}$, the general solution of the differential equation is

$$y = c_1 e^{-x/2} \cos \sqrt{4\lambda-1} x + c_2 e^{-x/2} \sin \sqrt{4\lambda-1} x.$$

The condition $y(0) = 0$ implies $c_1 = 0$ so $y = c_2 e^{-x/2} \sin \sqrt{4\lambda-1} x$. From

$$y(2) = c_2 e^{-1} \sin 2\sqrt{4\lambda-1} = 0$$

we see that the eigenvalues are determined by $2\sqrt{4\lambda-1} = n\pi$ for $n = 1, 2, 3, \dots$. Thus, the eigenvalues are $n^2\pi^2/4^2 + 1/4$ for $n = 1, 2, 3, \dots$, with corresponding eigenfunctions $e^{-x/2} \sin(n\pi x/2)$.

- (b) The self-adjoint form is

$$\frac{d}{dx}[e^x y'] + \lambda e^x y = 0.$$

12.5 Sturm-Liouville Problem

(c) An orthogonality relation is

$$\int_0^2 e^x \left(e^{-x/2} \sin \frac{m\pi}{2} x \right) \left(e^{-x/2} \cos \frac{n\pi}{2} x \right) dx = \int_0^2 \sin \frac{m\pi}{2} x \cos \frac{n\pi}{2} x dx = 0.$$

9. To obtain the self-adjoint form we note that an integrating factor is $(1/x)e^{\int(1-x)dx/x} = e^{-x}$. Thus, the differential equation is

$$xe^{-x}y'' + (1-x)e^{-x}y' + ne^{-x}y = 0$$

and the self-adjoint form is

$$\frac{d}{dx} [xe^{-x}y'] + ne^{-x}y = 0.$$

Identifying the weight function $p(x) = e^{-x}$ and noting that since $r(x) = xe^{-x}$, $r(0) = 0$ and $\lim_{x \rightarrow \infty} r(x) = 0$, we have the orthogonality relation

$$\int_0^\infty e^{-x} L_m(x) L_n(x) dx = 0, \quad m \neq n.$$

10. To obtain the self-adjoint form we note that an integrating factor is $e^{\int -2x dx} = e^{-x^2}$. Thus, the differential equation is

$$e^{-x^2}y'' - 2xe^{-x^2}y' + 2ne^{-x^2}y = 0$$

and the self-adjoint form is

$$\frac{d}{dx} [e^{-x^2}y'] + 2ne^{-x^2}y = 0.$$

Identifying the weight function $p(x) = e^{-x^2}$ and noting that since $r(x) = e^{-x^2}$, $\lim_{x \rightarrow -\infty} r(x) = \lim_{x \rightarrow \infty} r(x) = 0$, we have the orthogonality relation

$$\int_{-\infty}^\infty e^{-x^2} H_m(x) H_n(x) dx = 0, \quad m \neq n.$$

11. (a) The differential equation is

$$(1+x^2)y'' + 2xy' + \frac{\lambda}{1+x^2}y = 0.$$

Letting $x = \tan \theta$ we have $\theta = \tan^{-1} x$ and

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{d\theta} \frac{d\theta}{dx} = \frac{1}{1+x^2} \frac{dy}{d\theta} \\ \frac{d^2y}{dx^2} &= \frac{d}{dx} \left[\frac{1}{1+x^2} \frac{dy}{d\theta} \right] = \frac{1}{1+x^2} \left(\frac{d^2y}{d\theta^2} \frac{d\theta}{dx} \right) - \frac{2x}{(1+x^2)^2} \frac{dy}{d\theta} \\ &= \frac{1}{(1+x^2)^2} \frac{d^2y}{d\theta^2} - \frac{2x}{(1+x^2)^2} \frac{dy}{d\theta}. \end{aligned}$$

The differential equation can then be written in terms of $y(\theta)$ as

$$\begin{aligned} (1+x^2) \left[\frac{1}{(1+x^2)^2} \frac{d^2y}{d\theta^2} - \frac{2x}{(1+x^2)^2} \frac{dy}{d\theta} \right] + 2x \left[\frac{1}{1+x^2} \frac{dy}{d\theta} \right] + \frac{\lambda}{1+x^2} y \\ = \frac{1}{1+x^2} \frac{d^2y}{d\theta^2} + \frac{\lambda}{1+x^2} y = 0 \end{aligned}$$

or

$$\frac{d^2y}{d\theta^2} + \lambda y = 0.$$

The boundary conditions become $y(0) = y(\pi/4) = 0$. For $\lambda \leq 0$ the only solution of the boundary-value problem is $y = 0$. For $\lambda = \alpha^2 > 0$ the general solution of the differential equation is $y = c_1 \cos \alpha\theta + c_2 \sin \alpha\theta$. The condition $y(0) = 0$ implies $c_1 = 0$ so $y = c_2 \sin \alpha\theta$. Now the condition $y(\pi/4) = 0$ implies $c_2 \sin \alpha\pi/4 = 0$. For $c_2 \neq 0$ this condition will hold when $\alpha\pi/4 = n\pi$ or $\lambda = \alpha^2 = 16n^2$, where $n = 1, 2, 3, \dots$. These are the eigenvalues with corresponding eigenfunctions $\sin 4n\theta = \sin(4n \tan^{-1} x)$, for $n = 1, 2, 3, \dots$.

(b) An orthogonality relation is

$$\int_0^1 \frac{1}{x^2 + 1} \sin(4m \tan^{-1} x) \sin(4n \tan^{-1} x) dx = 0, \quad m \neq n.$$

12. (a) Letting $\lambda = \alpha^2$ the differential equation becomes $x^2 y'' + xy' + (\alpha^2 x^2 - 1)y = 0$. This is the parametric Bessel equation with $\nu = 1$. The general solution is

$$y = c_1 J_1(\alpha x) + c_2 Y_1(\alpha x).$$

Since Y is unbounded at 0 we must have $c_2 = 0$, so that $y = c_1 J_1(\alpha x)$. The condition $J_1(3\alpha) = 0$ defines the eigenvalues $\lambda_n = \alpha_n^2$ for $n = 1, 2, 3, \dots$. The corresponding eigenfunctions are $J_1(\alpha_n x)$.

- (b) Using a CAS or Table 5.1 in the text to solve $J_1(3\alpha) = 0$ we find $3\alpha_1 = 3.8317$, $3\alpha_2 = 7.0156$, $3\alpha_3 = 10.1735$, and $3\alpha_4 = 13.3237$. The corresponding eigenvalues are $\lambda_1 = \alpha_1^2 = 1.6313$, $\lambda_2 = \alpha_2^2 = 5.4687$, $\lambda_3 = \alpha_3^2 = 11.4999$, and $\lambda_4 = \alpha_4^2 = 19.7245$.

13. When $\lambda = 0$ the differential equation is $r(x)y'' + r'(x)y' = 0$. By inspection we see that $y = 1$ is a solution of the boundary-value problem. Thus, $\lambda = 0$ is an eigenvalue.

14. (a) An orthogonality relation is

$$\int_0^1 \cos x_m x \cos x_n x dx = 0$$

where $x_m \neq x_n$ are positive solutions of $\cot x = x$.

- (b) Referring to Problem 1 we use a CAS to compute

$$\int_0^1 (\cos 0.8603x)(\cos 3.4256x) dx = -1.8771 \times 10^{-6} \approx 0.$$

15. (a) An orthogonality relation is

$$\int_0^1 (x_m \cos x_m x - \sin x_m x)(x_n \cos x_n x - \sin x_n x) dx = 0$$

where $x_m \neq x_n$ are positive solutions of $\tan x = x$.

- (b) Referring to Problem 2 we use a CAS to compute

$$\int_0^1 (4.4934 \cos 4.4934x - \sin 4.4934x)(7.7253 \cos 7.7253x - \sin 7.7253x) dx = -2.5650 \times 10^{-4} \approx 0.$$

EXERCISES 12.6

Bessel and Legendre Series

- Identifying $b = 3$, we have $\alpha_1 = 1.2772$, $\alpha_2 = 2.3385$, $\alpha_3 = 3.3912$, and $\alpha_4 = 4.4412$.
- By (6) in the text $J'_0(2\alpha) = -J_1(2\alpha)$. Thus, $J'_0(2\alpha) = 0$ is equivalent to $J_1(2\alpha) = 0$. Then $\alpha_1 = 1.9159$, $\alpha_2 = 3.5078$, $\alpha_3 = 5.0867$, and $\alpha_4 = 6.6618$.
- The boundary condition indicates that we use (15) and (16) in the text. With $b = 2$ we obtain

$$\begin{aligned}
 c_i &= \frac{2}{4J_1^2(2\alpha_i)} \int_0^2 x J_0(\alpha_i x) dx && \boxed{t = \alpha_i x \quad dt = \alpha_i dx} \\
 &= \frac{1}{2J_1^2(2\alpha_i)} \cdot \frac{1}{\alpha_i^2} \int_0^{2\alpha_i} t J_0(t) dt \\
 &= \frac{1}{2\alpha_i^2 J_1^2(2\alpha_i)} \int_0^{2\alpha_i} \frac{d}{dt} [t J_1(t)] dt && \text{[From (5) in the text]} \\
 &= \frac{1}{2\alpha_i^2 J_1^2(2\alpha_i)} t J_1(t) \Big|_0^{2\alpha_i} \\
 &= \frac{1}{\alpha_i J_1(2\alpha_i)}.
 \end{aligned}$$

Thus

$$f(x) = \sum_{i=1}^{\infty} \frac{1}{\alpha_i J_1(2\alpha_i)} J_0(\alpha_i x).$$

- The boundary condition indicates that we use (19) and (20) in the text. With $b = 2$ we obtain

$$\begin{aligned}
 c_1 &= \frac{2}{4} \int_0^2 x dx = \frac{2}{4} \frac{x^2}{2} \Big|_0^2 = 1, \\
 c_i &= \frac{2}{4J_0^2(2\alpha_i)} \int_0^2 x J_0(\alpha_i x) dx && \boxed{t = \alpha_i x \quad dt = \alpha_i dx} \\
 &= \frac{1}{2J_0^2(2\alpha_i)} \cdot \frac{1}{\alpha_i^2} \int_0^{2\alpha_i} t J_0(t) dt \\
 &= \frac{1}{2\alpha_i^2 J_0^2(2\alpha_i)} \int_0^{2\alpha_i} \frac{d}{dt} [t J_1(t)] dt && \text{[From (5) in the text]} \\
 &= \frac{1}{2\alpha_i^2 J_0^2(2\alpha_i)} t J_1(t) \Big|_0^{2\alpha_i} \\
 &= \frac{J_1(2\alpha_i)}{\alpha_i J_0^2(2\alpha_i)}.
 \end{aligned}$$

Now since $J'_0(2\alpha_i) = 0$ is equivalent to $J_1(2\alpha_i) = 0$ we conclude $c_i = 0$ for $i = 2, 3, 4, \dots$. Thus the expansion of f on $0 < x < 2$ consists of a series with one nontrivial term:

$$f(x) = c_1 = 1.$$

5. The boundary condition indicates that we use (17) and (18) in the text. With $b = 2$ and $h = 1$ we obtain

$$\begin{aligned}
 c_i &= \frac{2\alpha_i^2}{(4\alpha_i^2 + 1)J_0^2(2\alpha_i)} \int_0^2 x J_0(\alpha_i x) dx & \boxed{t = \alpha_i x \quad dt = \alpha_i dx} \\
 &= \frac{2\alpha_i^2}{(4\alpha_i^2 + 1)J_0^2(2\alpha_i)} \cdot \frac{1}{\alpha_i^2} \int_0^{2\alpha_i} t J_0(t) dt \\
 &= \frac{2}{(4\alpha_i^2 + 1)J_0^2(2\alpha_i)} \int_0^{2\alpha_i} \frac{d}{dt} [t J_1(t)] dt & \text{[From (5) in the text]} \\
 &= \frac{2}{(4\alpha_i^2 + 1)J_0^2(2\alpha_i)} t J_1(t) \Big|_0^{2\alpha_i} \\
 &= \frac{4\alpha_i J_1(2\alpha_i)}{(4\alpha_i^2 + 1)J_0^2(2\alpha_i)}.
 \end{aligned}$$

Thus

$$f(x) = 4 \sum_{i=1}^{\infty} \frac{\alpha_i J_1(2\alpha_i)}{(4\alpha_i^2 + 1)J_0^2(2\alpha_i)} J_0(\alpha_i x).$$

6. Writing the boundary condition in the form

$$2J_0(2\alpha) + 2\alpha J_0'(2\alpha) = 0$$

we identify $b = 2$ and $h = 2$. Using (17) and (18) in the text we obtain

$$\begin{aligned}
 c_i &= \frac{2\alpha_i^2}{(4\alpha_i^2 + 4)J_0^2(2\alpha_i)} \int_0^2 x J_0(\alpha_i x) dx & \boxed{t = \alpha_i x \quad dt = \alpha_i dx} \\
 &= \frac{\alpha_i^2}{2(\alpha_i^2 + 1)J_0^2(2\alpha_i)} \cdot \frac{1}{\alpha_i^2} \int_0^{2\alpha_i} t J_0(t) dt \\
 &= \frac{1}{2(\alpha_i^2 + 1)J_0^2(2\alpha_i)} \int_0^{2\alpha_i} \frac{d}{dt} [t J_1(t)] dt & \text{[From (5) in the text]} \\
 &= \frac{1}{2(\alpha_i^2 + 1)J_0^2(2\alpha_i)} t J_1(t) \Big|_0^{2\alpha_i} \\
 &= \frac{\alpha_i J_1(2\alpha_i)}{(\alpha_i^2 + 1)J_0^2(2\alpha_i)}.
 \end{aligned}$$

Thus

$$f(x) = \sum_{i=1}^{\infty} \frac{\alpha_i J_1(2\alpha_i)}{(\alpha_i^2 + 1)J_0^2(2\alpha_i)} J_0(\alpha_i x).$$

7. The boundary condition indicates that we use (17) and (18) in the text. With $n = 1$, $b = 4$, and $h = 3$ we obtain

$$\begin{aligned}
 c_i &= \frac{2\alpha_i^2}{(16\alpha_i^2 - 1 + 9)J_1^2(4\alpha_i)} \int_0^4 x J_1(\alpha_i x) 5x dx & \boxed{t = \alpha_i x \quad dt = \alpha_i dx} \\
 &= \frac{5\alpha_i^2}{4(2\alpha_i^2 + 1)J_1^2(4\alpha_i)} \cdot \frac{1}{\alpha_i^3} \int_0^{4\alpha_i} t^2 J_1(t) dt \\
 &= \frac{5}{4\alpha_i(2\alpha_i^2 + 1)J_1^2(4\alpha_i)} \int_0^{4\alpha_i} \frac{d}{dt} [t^2 J_2(t)] dt & \text{[From (5) in the text]}
 \end{aligned}$$

12.6 Bessel and Legendre Series

$$\begin{aligned}
 &= \frac{5}{4\alpha_i(2\alpha_i^2 + 1)J_1^2(4\alpha_i)} t^2 J_2(t) \Big|_0^{4\alpha_i} \\
 &= \frac{20\alpha_i J_2(4\alpha_i)}{(2\alpha_i^2 + 1)J_1^2(4\alpha_i)}.
 \end{aligned}$$

Thus

$$f(x) = 20 \sum_{i=1}^{\infty} \frac{\alpha_i J_2(4\alpha_i)}{(2\alpha_i^2 + 1)J_1^2(4\alpha_i)} J_1(\alpha_i x).$$

8. The boundary condition indicates that we use (15) and (16) in the text. With $n = 2$ and $b = 1$ we obtain

$$\begin{aligned}
 c_1 &= \frac{2}{J_3^2(\alpha_i)} \int_0^1 x J_2(\alpha_i x) x^2 dx && \boxed{t = \alpha_i x \quad dt = \alpha_i dx} \\
 &= \frac{2}{J_3^2(\alpha_i)} \cdot \frac{1}{\alpha_i^4} \int_0^{\alpha_i} t^3 J_2(t) dt \\
 &= \frac{2}{\alpha_i^4 J_3^2(\alpha_i)} \int_0^{\alpha_i} \frac{d}{dt} [t^3 J_3(t)] dt && [\text{From (5) in the text}] \\
 &= \frac{2}{\alpha_i^4 J_3^2(\alpha_i)} t^3 J_3(t) \Big|_0^{\alpha_i} \\
 &= \frac{2}{\alpha_i J_3(\alpha_i)}.
 \end{aligned}$$

Thus

$$f(x) = 2 \sum_{i=1}^{\infty} \frac{1}{\alpha_i J_3(\alpha_i)} J_2(\alpha_i x).$$

9. The boundary condition indicates that we use (19) and (20) in the text. With $b = 3$ we obtain

$$\begin{aligned}
 c_1 &= \frac{2}{9} \int_0^3 x x^2 dx = \frac{2}{9} \frac{x^4}{4} \Big|_0^3 = \frac{9}{2}, \\
 c_i &= \frac{2}{9J_0^2(3\alpha_i)} \int_0^3 x J_0(\alpha_i x) x^2 dx && \boxed{t = \alpha_i x \quad dt = \alpha_i dx} \\
 &= \frac{2}{9J_0^2(3\alpha_i)} \cdot \frac{1}{\alpha_i^4} \int_0^{3\alpha_i} t^3 J_0(t) dt \\
 &= \frac{2}{9\alpha_i^4 J_0^2(3\alpha_i)} \int_0^{3\alpha_i} t^2 \frac{d}{dt} [t J_1(t)] dt && \boxed{\begin{array}{ll} u = t^2 & dv = \frac{d}{dt} [t J_1(t)] dt \\ du = 2t dt & v = t J_1(t) \end{array}} \\
 &= \frac{2}{9\alpha_i^4 J_0^2(3\alpha_i)} \left(t^3 J_1(t) \Big|_0^{3\alpha_i} - 2 \int_0^{3\alpha_i} t^2 J_1(t) dt \right).
 \end{aligned}$$

With $n = 0$ in equation (6) in the text we have $J'_0(x) = -J_1(x)$, so the boundary condition $J'_0(3\alpha_i) = 0$ implies $J_1(3\alpha_i) = 0$. Then

$$\begin{aligned} c_i &= \frac{2}{9\alpha_i^4 J_0^2(3\alpha_i)} \left(-2 \int_0^{3\alpha_i} \frac{d}{dt} [t^2 J_2(t)] dt \right) = \frac{2}{9\alpha_i^4 J_0^2(3\alpha_i)} \left(-2t^2 J_2(t) \Big|_0^{3\alpha_i} \right) \\ &= \frac{2}{9\alpha_i^4 J_0^2(3\alpha_i)} [-18\alpha_i^2 J_2(3\alpha_i)] = \frac{-4J_2(3\alpha_i)}{\alpha_i^2 J_0^2(3\alpha_i)}. \end{aligned}$$

Thus

$$f(x) = \frac{9}{2} - 4 \sum_{i=1}^{\infty} \frac{J_2(3\alpha_i)}{\alpha_i^2 J_0^2(3\alpha_i)} J_0(\alpha_i x).$$

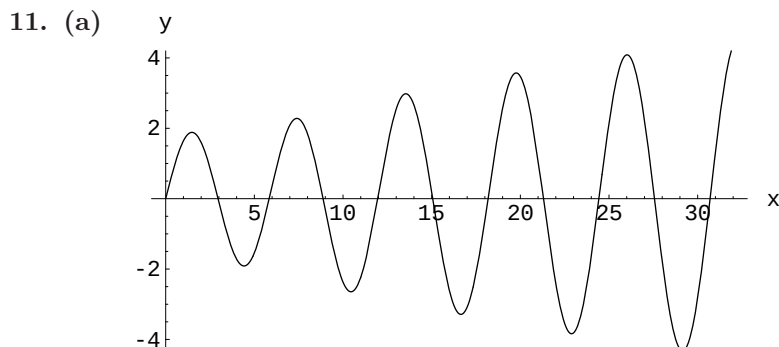
10. The boundary condition indicates that we use (15) and (16) in the text. With $b = 1$ it follows that

$$\begin{aligned} c_i &= \frac{2}{J_1^2(\alpha_i)} \int_0^1 x(1-x^2) J_0(\alpha_i x) dx \\ &= \frac{2}{J_1^2(\alpha_i)} \left[\int_0^1 x J_0(\alpha_i x) dx - \int_0^1 x^3 J_0(\alpha_i x) dx \right] \\ &= \frac{2}{J_1^2(\alpha_i)} \left[\frac{1}{\alpha_i^2} \int_0^{\alpha_i} t J_0(t) dt - \frac{1}{\alpha_i^4} \int_0^{\alpha_i} t^3 J_0(t) dt \right] \\ &= \frac{2}{J_1^2(\alpha_i)} \left[\frac{1}{\alpha_i^2} \int_0^{\alpha_i} \frac{d}{dt} [t J_1(t)] dt - \frac{1}{\alpha_i^4} \int_0^{\alpha_i} t^2 \frac{d}{dt} [t J_1(t)] dt \right] \\ &= \frac{2}{J_1^2(\alpha_i)} \left[\frac{1}{\alpha_i^2} t J_1(t) \Big|_0^{\alpha_i} - \frac{1}{\alpha_i^4} \left(t^3 J_1(t) \Big|_0^{\alpha_i} - 2 \int_0^{\alpha_i} t^2 J_1(t) dt \right) \right] \\ &= \frac{2}{J_1^2(\alpha_i)} \left[\frac{J_1(\alpha_i)}{\alpha_i} - \frac{J_1(\alpha_i)}{\alpha_i} + \frac{2}{\alpha_i^4} \int_0^{\alpha_i} \frac{d}{dt} [t^2 J_2(t)] dt \right] \\ &= \frac{2}{J_1^2(\alpha_i)} \left[\frac{2}{\alpha_i^4} t^2 J_2(t) \Big|_0^{\alpha_i} \right] = \frac{4J_2(\alpha_i)}{\alpha_i^2 J_1^2(\alpha_i)}. \end{aligned}$$

$$\begin{aligned} t &= \alpha_i x & dt &= \alpha_i dx \\ u &= t^2 & dv &= \frac{d}{dt} [t J_1(t)] dt \\ du &= 2t dt & v &= t J_1(t) \end{aligned}$$

Thus

$$f(x) = 4 \sum_{i=1}^{\infty} \frac{J_2(\alpha_i)}{\alpha_i^2 J_1^2(\alpha_i)} J_0(\alpha_i x).$$



12.6 Bessel and Legendre Series

(b) Using **FindRoot** in *Mathematica* we find the roots $x_1 = 2.9496$, $x_2 = 5.8411$, $x_3 = 8.8727$, $x_4 = 11.9561$, and $x_5 = 15.0624$.

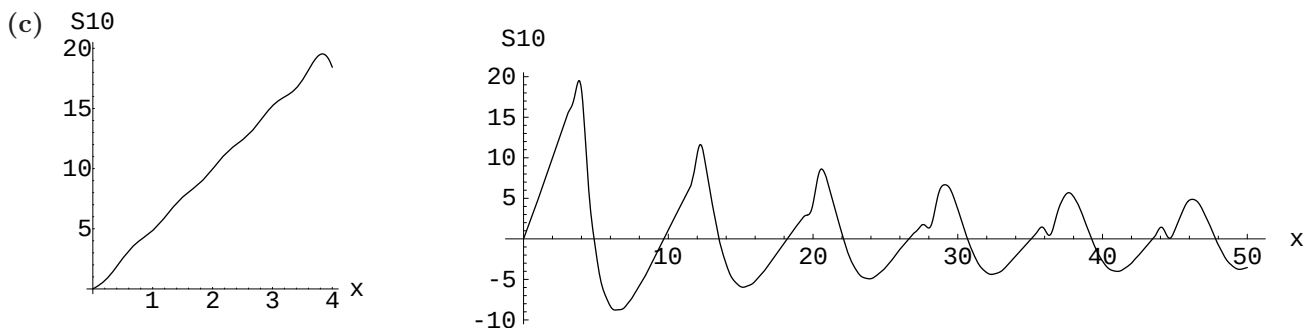
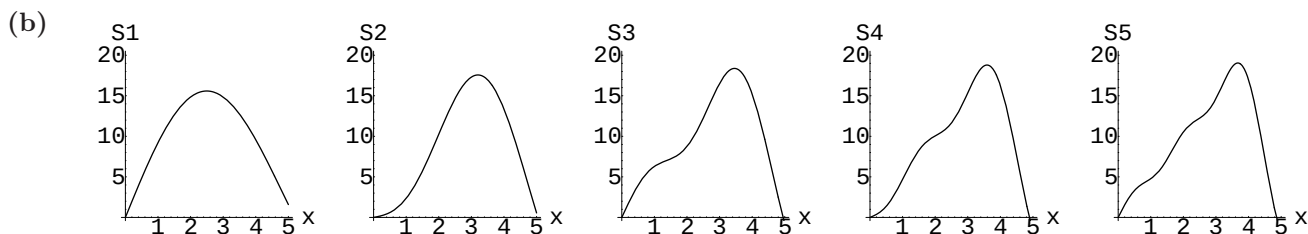
(c) Dividing the roots in part (b) by 4 we find the eigenvalues $\alpha_1 = 0.7374$, $\alpha_2 = 1.4603$, $\alpha_3 = 2.2182$, $\alpha_4 = 2.9890$, and $\alpha_5 = 3.7656$.

(d) The next five eigenvalues are $\alpha_6 = 4.5451$, $\alpha_7 = 5.3263$, $\alpha_8 = 6.1085$, $\alpha_9 = 6.8915$, and $\alpha_{10} = 7.6749$.

12. (a) From Problem 7, the coefficients of the Fourier-Bessel series are

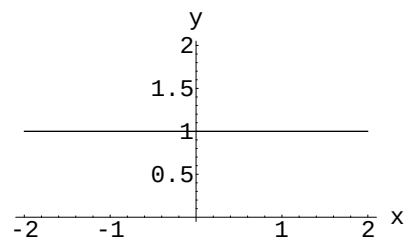
$$c_i = \frac{20\alpha_i J_2(4\alpha_i)}{(2\alpha_i^2 + 1)J_1^2(4\alpha_i)}.$$

Using a CAS we find $c_1 = 26.7896$, $c_2 = -12.4624$, $c_3 = 7.1404$, $c_4 = -4.68705$, and $c_5 = 3.35619$.



13. Since f is expanded as a series of Bessel functions, $J_1(\alpha_i x)$ and J_1 is an odd function, the series should represent an odd function.

14. (a) Since J_0 is an even function, a series expansion of a function defined on $(0, 2)$ would converge to the even extension of the function on $(-2, 0)$.



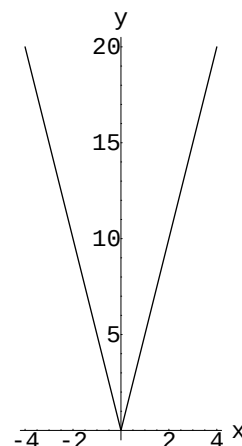
- (b) In Section 5.3 we saw that $J'_2(x) = 2J_2(x)/x - J_3(x)$. Since J_2 is even and J_3 is odd we see that

$$\begin{aligned} J'_2(-x) &= 2J_2(-x)/(-x) - J_3(-x) \\ &= -2J_2(x)/x + J_3(x) = -J'_2(x), \end{aligned}$$

so that J'_2 is an odd function. Now, if $f(x) = 3J_2(x) + 2xJ'_2(x)$, we see that

$$\begin{aligned} f(-x) &= 3J_2(-x) - 2xJ'_2(-x) \\ &= 3J_2(x) + 2xJ'_2(x) = f(x), \end{aligned}$$

so that f is an even function. Thus, a series expansion of a function defined on $(0, 4)$ would converge to the even extension of the function on $(-4, 0)$.



15. We compute

$$c_0 = \frac{1}{2} \int_0^1 x P_0(x) dx = \frac{1}{2} \int_0^1 x dx = \frac{1}{4}$$

$$c_1 = \frac{3}{2} \int_0^1 x P_1(x) dx = \frac{3}{2} \int_0^1 x^2 dx = \frac{1}{2}$$

$$c_2 = \frac{5}{2} \int_0^1 x P_2(x) dx = \frac{5}{2} \int_0^1 \frac{1}{2}(3x^3 - x) dx = \frac{5}{16}$$

$$c_3 = \frac{7}{2} \int_0^1 x P_3(x) dx = \frac{7}{2} \int_0^1 \frac{1}{2}(5x^4 - 3x^2) dx = 0$$

$$c_4 = \frac{9}{2} \int_0^1 x P_4(x) dx = \frac{9}{2} \int_0^1 \frac{1}{8}(35x^5 - 30x^3 + 3x) dx = -\frac{3}{32}$$

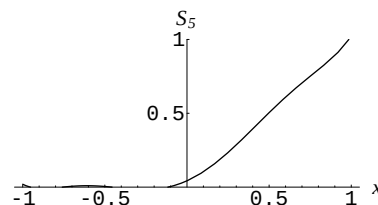
$$c_5 = \frac{11}{2} \int_0^1 x P_5(x) dx = \frac{11}{2} \int_0^1 \frac{1}{8}(63x^6 - 70x^4 + 15x^2) dx = 0$$

$$c_6 = \frac{13}{2} \int_0^1 x P_6(x) dx = \frac{13}{2} \int_0^1 \frac{1}{16}(231x^7 - 315x^5 + 105x^3 - 5x) dx = \frac{13}{256}.$$

Thus

$$f(x) = \frac{1}{4}P_0(x) + \frac{1}{2}P_1(x) + \frac{5}{16}P_2(x) - \frac{3}{32}P_4(x) + \frac{13}{256}P_6(x) + \cdots.$$

The figure above is the graph of $S_5(x) = \frac{1}{4}P_0(x) + \frac{1}{2}P_1(x) + \frac{5}{16}P_2(x) - \frac{3}{32}P_4(x) + \frac{13}{256}P_6(x)$.



12.6 Bessel and Legendre Series

16. We compute

$$c_0 = \frac{1}{2} \int_{-1}^1 e^x P_0(x) dx = \frac{1}{2} \int_{-1}^1 e^x dx = \frac{1}{2}(e - e^{-1})$$

$$c_1 = \frac{3}{2} \int_{-1}^1 e^x P_1(x) dx = \frac{3}{2} \int_{-1}^1 x e^x dx = 3e^{-1}$$

$$\begin{aligned} c_2 &= \frac{5}{2} \int_{-1}^1 e^x P_2(x) dx = \frac{5}{2} \int_{-1}^1 \frac{1}{2}(3x^2 e^x - e^x) dx \\ &= \frac{5}{2}(e - 7e^{-1}) \end{aligned}$$

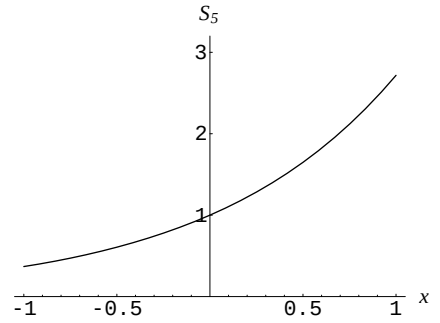
$$c_3 = \frac{7}{2} \int_{-1}^1 e^x P_3(x) dx = \frac{7}{2} \int_{-1}^1 \frac{1}{2}(5x^3 e^x - 3x e^x) dx = \frac{7}{2}(-5e + 37e^{-1})$$

$$c_4 = \frac{9}{2} \int_{-1}^1 e^x P_4(x) dx = \frac{9}{2} \int_{-1}^1 \frac{1}{8}(35x^4 e^x - 30x^2 e^x + 3e^x) dx = \frac{9}{2}(36e - 266e^{-1}).$$

Thus

$$\begin{aligned} f(x) &= \frac{1}{2}(e - e^{-1})P_0(x) + 3e^{-1}P_1(x) + \frac{5}{2}(e - 7e^{-1})P_2(x) \\ &\quad + \frac{7}{2}(-5e + 37e^{-1})P_3(x) + \frac{9}{2}(36e - 266e^{-1})P_4(x) + \cdots \end{aligned}$$

The figure above is the graph of $S_5(x)$.



17. Using $\cos^2 \theta = \frac{1}{2}(\cos 2\theta + 1)$ we have

$$\begin{aligned} P_2(\cos \theta) &= \frac{1}{2}(3 \cos^2 \theta - 1) = \frac{3}{2} \cos^2 \theta - \frac{1}{2} \\ &= \frac{3}{4}(\cos 2\theta + 1) - \frac{1}{2} = \frac{3}{4} \cos 2\theta + \frac{1}{4} = \frac{1}{4}(3 \cos 2\theta + 1). \end{aligned}$$

18. From Problem 17 we have

$$P_2(\cos \theta) = \frac{1}{4}(3 \cos 2\theta + 1) \quad \text{or} \quad \cos 2\theta = \frac{4}{3}P_2(\cos \theta) - \frac{1}{3}.$$

Then, using $P_0(\cos \theta) = 1$,

$$\begin{aligned} F(\theta) &= 1 - \cos 2\theta = 1 - \left[\frac{4}{3}P_2(\cos \theta) - \frac{1}{3} \right] \\ &= \frac{4}{3} - \frac{4}{3}P_2(\cos \theta) = \frac{4}{3}P_0(\cos \theta) - \frac{4}{3}P_2(\cos \theta). \end{aligned}$$

19. If f is an even function on $(-1, 1)$ then

$$\int_{-1}^1 f(x) P_{2n}(x) dx = 2 \int_0^1 f(x) P_{2n}(x) dx$$

and

$$\int_{-1}^1 f(x) P_{2n+1}(x) dx = 0.$$

Thus

$$\begin{aligned} c_{2n} &= \frac{2(2n)+1}{2} \int_{-1}^1 f(x)P_{2n}(x) dx = \frac{4n+1}{2} \left(2 \int_0^1 f(x)P_{2n}(x) dx \right) \\ &= (4n+1) \int_0^1 f(x)P_{2n}(x) dx, \end{aligned}$$

$c_{2n+1} = 0$, and

$$f(x) = \sum_{n=0}^{\infty} c_{2n} P_{2n}(x).$$

20. If f is an odd function on $(-1, 1)$ then

$$\int_{-1}^1 f(x)P_{2n}(x) dx = 0$$

and

$$\int_{-1}^1 f(x)P_{2n+1}(x) dx = 2 \int_0^1 f(x)P_{2n+1}(x) dx.$$

Thus

$$\begin{aligned} c_{2n+1} &= \frac{2(2n+1)+1}{2} \int_{-1}^1 f(x)P_{2n+1}(x) dx = \frac{4n+3}{2} \left(2 \int_0^1 f(x)P_{2n+1}(x) dx \right) \\ &= (4n+3) \int_0^1 f(x)P_{2n+1}(x) dx, \end{aligned}$$

$c_{2n} = 0$, and

$$f(x) = \sum_{n=0}^{\infty} c_{2n+1} P_{2n+1}(x).$$

21. From (26) in Problem 19 in the text we find

$$c_0 = \int_0^1 xP_0(x) dx = \int_0^1 x dx = \frac{1}{2},$$

$$c_2 = 5 \int_0^1 xP_2(x) dx = 5 \int_0^1 \frac{1}{2}(3x^3 - x) dx = \frac{5}{8},$$

$$c_4 = 9 \int_0^1 xP_4(x) dx = 9 \int_0^1 \frac{1}{8}(35x^5 - 30x^3 + 3x) dx = -\frac{3}{16},$$

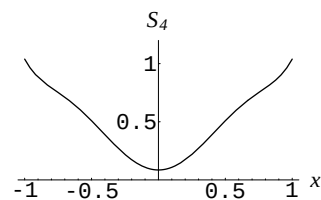
and

$$c_6 = 13 \int_0^1 xP_6(x) dx = 13 \int_0^1 \frac{1}{16}(231x^7 - 315x^5 + 105x^3 - 5x) dx = \frac{13}{128}.$$

Hence, from (25) in the text,

$$f(x) = \frac{1}{2}P_0(x) + \frac{5}{8}P_2(x) - \frac{3}{16}P_4(x) + \frac{13}{128}P_6 + \cdots.$$

On the interval $-1 < x < 1$ this series represents the function $f(x) = |x|$.



12.6 Bessel and Legendre Series

22. From (28) in Problem 20 in the text we find

$$c_1 = 3 \int_0^1 P_1(x) dx = 3 \int_0^1 x dx = \frac{3}{2},$$

$$c_3 = 7 \int_0^1 P_3(x) dx = 7 \int_0^1 \frac{1}{2} (5x^3 - 3x) dx = -\frac{7}{8},$$

$$c_5 = 11 \int_0^1 P_5(x) dx = 11 \int_0^1 \frac{1}{8} (63x^5 - 70x^3 + 15x) dx = \frac{11}{16}$$

and

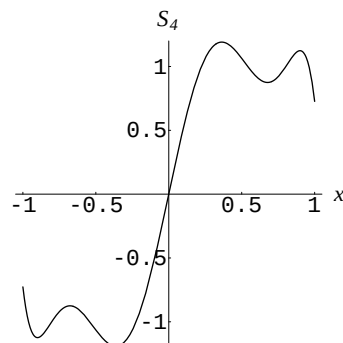
$$c_7 = 15 \int_0^1 P_7(x) dx = 15 \int_0^1 \frac{1}{16} (429x^7 - 693x^5 + 315x^3 - 35x) dx = -\frac{75}{128}.$$

Hence, from (27) in the text,

$$f(x) = \frac{3}{2}P_1(x) - \frac{7}{8}P_3(x) + \frac{11}{16}P_5(x) - \frac{75}{128}P_7(x) + \cdots.$$

On the interval $-1 < x < 1$ this series represents the odd function

$$f(x) = \begin{cases} -1, & -1 < x < 0 \\ 1, & 0 < x < 1. \end{cases}$$



23. Since there is a Legendre polynomial of any specified degree, every polynomial can be represented as a finite linear combination of Legendre polynomials.

24. We want to express both x^2 and x^3 as linear combinations of $P_0(x) = 1$, $P_1(x) = x$, $P_2(x) = \frac{1}{2}(3x^2 - 1)$, and $P_3(x) = \frac{1}{2}(5x^3 - 3x)$. Setting

$$x^2 = c_0 P_0(x) + c_1 P_1(x) + c_2 P_2(x) = c_0 + c_1 x + c_2 \left[\frac{1}{2}(3x^2 - 1) \right] = \left(c_0 - \frac{3}{2}c_2 \right) + c_1 x + \frac{3}{2}c_2 x^2,$$

we obtain the system

$$\begin{aligned} c_0 - \frac{3}{2}c_2 &= 0 \\ c_1 &= 0 \\ \frac{3}{2}c_2 &= 1. \end{aligned}$$

The solution is $c_0 = \frac{1}{3}$, $c_1 = 0$, $c_2 = \frac{2}{3}$. Thus, $x^2 = \frac{1}{3}P_0(x) + \frac{2}{3}P_2(x)$. Setting

$$\begin{aligned} x^3 &= c_0 P_0(x) + c_1 P_1(x) + c_2 P_2(x) + c_3 P_3(x) = c_0 + c_1 x + c_2 \left[\frac{1}{2}(3x^2 - 1) \right] + c_3 \left[\frac{1}{2}(5x^3 - 3x) \right] \\ &= \left(c_0 - \frac{1}{2}c_2 \right) + \left(c_1 - \frac{3}{2}c_3 \right) x + \frac{3}{2}c_2 x^2 + \frac{5}{2}c_3 x^3, \end{aligned}$$

we obtain the system

$$\begin{aligned} c_0 - \frac{1}{2}c_2 &= 0 \\ c_1 - \frac{3}{2}c_3 &= 0 \\ \frac{3}{2}c_2 &= 0 \\ \frac{5}{2}c_3 &= 1. \end{aligned}$$

The solution is $c_0 = 0$, $c_1 = \frac{3}{5}$, $c_2 = 0$, $c_3 = \frac{2}{5}$. Thus $x^3 = \frac{3}{5}P_1(x) + \frac{2}{5}P_3(x)$.

CHAPTER 12 REVIEW EXERCISES

1. True, since $\int_{-\pi}^{\pi} (x^2 - 1)x^5 dx = 0$
2. Even, since if f and g are odd then $h(-x) = f(-x)g(-x) = -f(x)[-g(x)] = f(x)g(x) = h(x)$
3. cosine, since f is even
4. True
5. False; the Sturm-Liouville problem,

$$\frac{d}{dx}[r(x)y'] + \lambda p(x)y = 0, \quad y'(a) = 0, \quad y'(b) = 0,$$

on the interval $[a, b]$, has eigenvalue $\lambda = 0$.

6. Periodically extending the function we see that at $x = -1$ the function converges to $\frac{1}{2}(-1 + 0) = -\frac{1}{2}$; at $x = 0$ it converges to $\frac{1}{2}(0 + 1) = \frac{1}{2}$, and at $x = 1$ it converges to $\frac{1}{2}(-1 + 0) = -\frac{1}{2}$.
7. The Fourier series will converge to 1, the cosine series to 1, and the sine series to 0 at $x = 0$. Respectively, this is because the rule $(x^2 + 1)$ defining $f(x)$ determines a continuous function on $(-3, 3)$, the even extension of f to $(-3, 0)$ is continuous at 0, and the odd extension of f to $(-3, 0)$ approaches -1 as x approaches 0 from the left.
8. $\cos 5x$, since the general solution is $y = c_1 \cos \alpha x + c_2 \sin \alpha x$ and $y'(0) = 0$ implies $c_2 = 0$.
9. True, since $\int_0^1 P_{2m}(x)P_{2n}(x) dx = \frac{1}{2} \int_{-1}^1 P_{2m}(x)P_{2n}(x) dx = 0$ when $m \neq n$.
10. Since $P_n(x)$ is orthogonal to $P_0(x) = 1$ for $n > 0$,

$$\int_{-1}^1 P_n(x) dx = \int_{-1}^1 P_0(x)P_n(x) dx = 0.$$

11. We know from a half-angle formula in trigonometry that $\cos^2 x = \frac{1}{2} + \frac{1}{2} \cos 2x$, which is a cosine series.
12. (a) For $m \neq n$

$$\int_0^L \sin \frac{(2n+1)\pi}{2L} x \sin \frac{(2m+1)\pi}{2L} x dx = \frac{1}{2} \int_0^L \left(\cos \frac{n-m}{L} \pi x - \cos \frac{n+m+1}{L} \pi x \right) dx = 0.$$

(b) From

$$\int_0^L \sin^2 \frac{(2n+1)\pi}{2L} x dx = \int_0^L \left(\frac{1}{2} - \frac{1}{2} \cos \frac{(2n+1)\pi}{L} x \right) dx = \frac{L}{2}$$

we see that

$$\left\| \sin \frac{(2n+1)\pi}{2L} x \right\| = \sqrt{\frac{L}{2}}.$$

13. Since

$$a_0 = \int_{-1}^0 (-2x) dx = 1,$$

$$a_n = \int_{-1}^0 (-2x) \cos n\pi x dx = \frac{2}{n^2 \pi^2} [(-1)^n - 1],$$

and

CHAPTER 12 REVIEW EXERCISES

$$b_n = \int_{-1}^0 (-2x) \sin n\pi x \, dx = \frac{4}{n\pi}(-1)^n$$

for $n = 1, 2, 3, \dots$ we have

$$f(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \left(\frac{2}{n^2\pi^2} [(-1)^n - 1] \cos n\pi x + \frac{4}{n\pi} (-1)^n \sin n\pi x \right).$$

14. Since

$$a_0 = \int_{-1}^1 (2x^2 - 1) \, dx = -\frac{2}{3},$$

$$a_n = \int_{-1}^1 (2x^2 - 1) \cos n\pi x \, dx = \frac{8}{n^2\pi^2}(-1)^n,$$

and

$$b_n = \int_{-1}^1 (2x^2 - 1) \sin n\pi x \, dx = 0$$

for $n = 1, 2, 3, \dots$ we have

$$f(x) = -\frac{1}{3} + \sum_{n=1}^{\infty} \frac{8}{n^2\pi^2}(-1)^n \cos n\pi x.$$

15. Since

$$a_0 = \frac{2}{1} \int_0^1 e^x \, dx = 2(e - 1)$$

and

$$a_n = \frac{2}{1} \int_0^1 e^x \cos n\pi x \, dx = \frac{2}{1 + n^2\pi^2} [e(-1)^n - 1],$$

for $n = 1, 2, 3, \dots$, we have the cosine series

$$f(x) = e - 1 + 2 \sum_{n=1}^{\infty} \frac{e(-1)^n - 1}{1 + n^2\pi^2} \cos n\pi x.$$

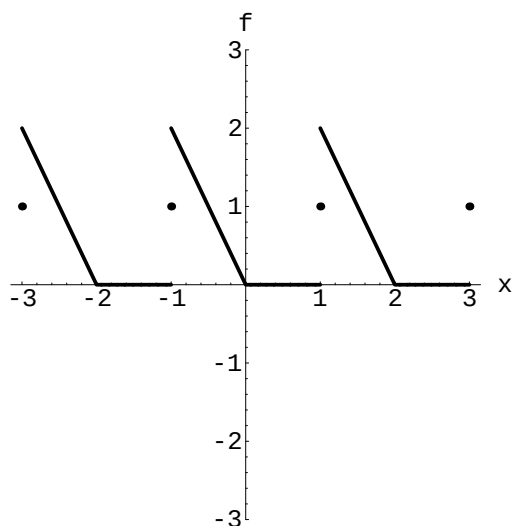
Since

$$b_n = \frac{2}{1} \int_0^1 e^x \sin n\pi x \, dx = \frac{2n\pi}{1 + n^2\pi^2} [1 - e(-1)^n],$$

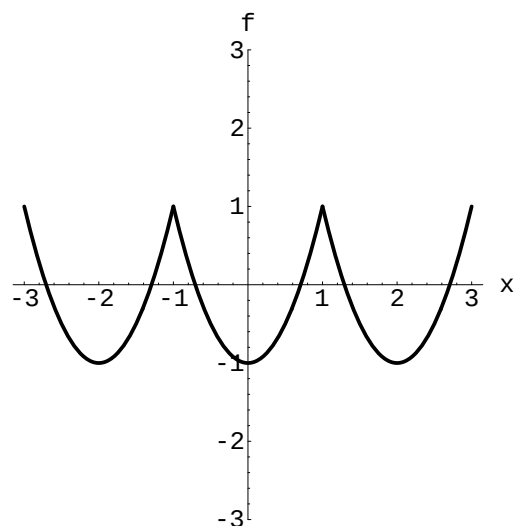
for $n = 1, 2, 3, \dots$, we have the sine series

$$f(x) = 2\pi \sum_{n=1}^{\infty} \frac{n[1 - e(-1)^n]}{1 + n^2\pi^2} \sin n\pi x.$$

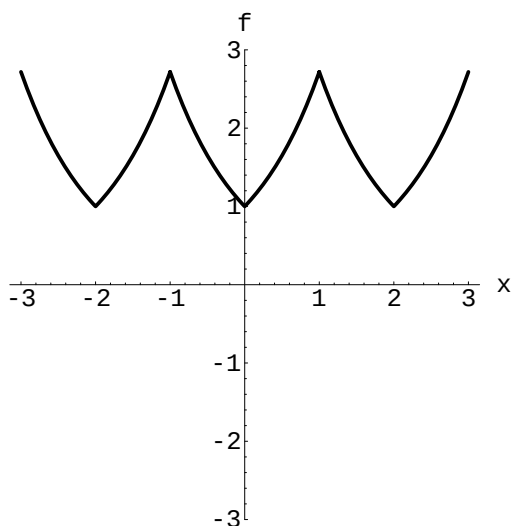
16.



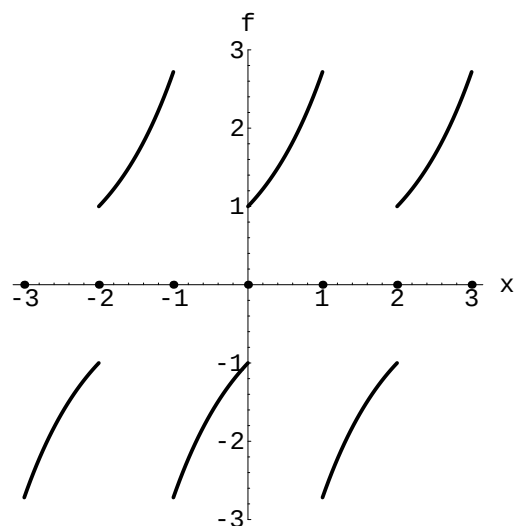
$$f(x) = |x| - x, \quad -1 < x < 1$$



$$f(x) = 2x^2 - 1, \quad -1 < x < 1$$



$$f(x) = \begin{cases} e^{-x}, & -1 < x < 0 \\ e^x, & 0 < x < 1 \end{cases}$$



$$f(x) = \begin{cases} -e^{-x}, & -1 < x < 0 \\ e^x, & 0 < x < 1 \end{cases}$$

 17. For $\lambda = \alpha^2 > 0$ a general solution of the given differential equation is

$$y = c_1 \cos(3\alpha \ln x) + c_2 \sin(3\alpha \ln x)$$

and

$$y' = -\frac{3c_1\alpha}{x} \sin(3\alpha \ln x) + \frac{3c_2\alpha}{x} \cos(3\alpha \ln x).$$

 Since $\ln 1 = 0$, the boundary condition $y'(1) = 0$ implies $c_2 = 0$. Therefore

$$y = c_1 \cos(3\alpha \ln x).$$

Using $\ln e = 1$ we find that $y(e) = 0$ implies $c_1 \cos 3\alpha = 0$ or $3\alpha = (2n - 1)\pi/2$, for $n = 1, 2, 3, \dots$. The eigenvalues are $\lambda = \alpha^2 = (2n - 1)^2\pi^2/36$ with corresponding eigenfunctions $\cos[(2n - 1)\pi(\ln x)/2]$ for $n = 1, 2, 3, \dots$.

CHAPTER 12 REVIEW EXERCISES

18. To obtain the self-adjoint form of the differential equation in Problem 17 we note that an integrating factor is $(1/x^2)e^{\int dx/x} = 1/x$. Thus the weight function is $1/x$ and an orthogonality relation is

$$\int_1^e \frac{1}{x} \cos\left(\frac{2n-1}{2}\pi \ln x\right) \cos\left(\frac{2m-1}{2}\pi \ln x\right) dx = 0, \quad m \neq n.$$

19. Since the coefficient of y in the differential equation is n^2 , the weight function is the integrating factor

$$\frac{1}{a(x)} e^{\int (b/a) dx} = \frac{1}{1-x^2} e^{\int -\frac{x}{1-x^2} dx} = \frac{1}{1-x^2} e^{\frac{1}{2} \ln(1-x^2)} = \frac{\sqrt{1-x^2}}{1-x^2} = \frac{1}{\sqrt{1-x^2}}$$

on the interval $[-1, 1]$. The orthogonality relation is

$$\int_{-1}^1 \frac{1}{\sqrt{1-x^2}} T_m(x) T_n(x) dx = 0, \quad m \neq n.$$

20. Expanding in a full Fourier series we have

$$\begin{aligned} a_0 &= \frac{1}{2} \left(\int_0^2 x dx + \int_2^4 2 dx \right) = 3 \\ a_n &= \frac{1}{2} \left(\int_0^2 x \cos \frac{n\pi x}{2} dx + \int_2^4 2 \cos \frac{n\pi x}{2} dx \right) = 2 \frac{(-1)^n - 1}{n^2 \pi^2} \\ b_n &= \frac{1}{2} \left(\int_0^2 x \sin \frac{n\pi x}{2} dx + \int_2^4 2 \sin \frac{n\pi x}{2} dx \right) = 4 \frac{-1}{n\pi} \end{aligned}$$

so

$$f(x) = \frac{3}{2} + 2 \sum_{n=1}^{\infty} \left(\frac{(-1)^n - 1}{n^2 \pi^2} \cos \frac{n\pi x}{2} - \frac{2}{n\pi} \sin \frac{n\pi x}{2} \right).$$

21. The boundary condition indicates that we use (15) and (16) of Section 12.6 in the text. With $b = 4$ we obtain

$$\begin{aligned} c_i &= \frac{2}{16J_1^2(4\alpha_i)} \int_0^4 x J_0(\alpha_i x) f(x) dx \\ &= \frac{1}{8J_1^2(4\alpha_i)} \int_0^2 x J_0(\alpha_i x) dx \\ &= \frac{1}{8J_1^2(4\alpha_i)} \cdot \frac{1}{\alpha_i^2} \int_0^{2\alpha_i} t J_0(t) dt \\ &= \frac{1}{8J_1^2(4\alpha_i)} \int_0^{2\alpha_i} \frac{d}{dt} [t J_1(t)] dt && \boxed{t = \alpha_i x \quad dt = \alpha_i dx} \\ &= \frac{1}{8J_1^2(4\alpha_i)} t J_1(t) \Big|_0^{2\alpha_i} = \frac{J_1(2\alpha_i)}{4\alpha_i J_1^2(4\alpha_i)}. \end{aligned}$$

[From (5) in 12.6 in the text]

Thus

$$f(x) = \frac{1}{4} \sum_{i=1}^{\infty} \frac{J_1(2\alpha_i)}{\alpha_i J_1^2(4\alpha_i)} J_0(\alpha_i x).$$

22. Since $f(x) = x^4$ is a polynomial in x , an expansion of f in Legendre polynomials in x must terminate with the term having the same degree as f . Using the fact that $x^4 P_1(x)$ and $x^4 P_3(x)$ are odd functions, we see immediately that $c_1 = c_3 = 0$. Now

$$\begin{aligned} c_0 &= \frac{1}{2} \int_{-1}^1 x^4 P_0(x) dx = \frac{1}{2} \int_{-1}^1 x^4 dx = \frac{1}{5} \\ c_2 &= \frac{5}{2} \int_{-1}^1 x^4 P_2(x) dx = \frac{5}{2} \int_{-1}^1 \frac{1}{2}(3x^6 - x^4) dx = \frac{4}{7} \\ c_4 &= \frac{9}{2} \int_{-1}^1 x^4 P_4(x) dx = \frac{9}{2} \int_{-1}^1 \frac{1}{8}(35x^8 - 30x^6 + 3x^4) dx = \frac{8}{35}. \end{aligned}$$

Thus

$$f(x) = \frac{1}{5}P_0(x) + \frac{4}{7}P_2(x) + \frac{8}{35}P_4(x).$$